

CMB Comparison-Map Closure in CPTG

Real Planck and WMAP Temperature-Map Tests

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Abstract

The cosmic microwave background is not merely a sky image; it is a measured record of the large-scale angular organization of the early Universe. This paper places a locked geometric construction of Curvature Polarization Transport Gravity (CPTG) directly into that observational field. The CPTG construction is translated into the measured CMB harmonic coordinate, carried across hundreds of multipoles and their spherical-harmonic modes, and reconstructed in real Planck and WMAP temperature-map space without sector-specific theory retuning.

The result is a field-level geometric closure. In the primary Planck comparison, the reconstructed CPTG field reaches an observed-field correlation of 0.983434, while across the five principal Planck component products the mean CPTG–Planck residual-RMS separation is only $0.023518\ \mu\text{K}$. The same narrow separation survives Planck split maps, WMAP low- ℓ bands, smoothing controls, masks, and sky-fraction changes. Under the identical observational-coordinate translation, generic flat, power-law, and shuffled angular-power envelopes fail strongly; shuffled spectra produce residual RMS near $95.46\ \mu\text{K}$, compared with approximately $18.23\ \mu\text{K}$ for Planck and $18.25\ \mu\text{K}$ for CPTG.

The practical significance is that a fixed geometric formula can be taken from its native theoretical description and expressed directly in the measured angular structure of the real CMB sky without being rewritten to fit each dataset. The comparison map is therefore the visible endpoint of a deeper result: the CPTG geometry survives translation into one of the richest observational fields available in cosmology while retaining the angular organization required by the data.

1 Motivation

The cosmic microwave background is usually treated through angular-power spectra, likelihoods, nuisance models, and component-separation pipelines. A direct comparison between a native geometric theory and CMB data therefore requires an observational interface. In CPTG, the locked native geometric-pi branch is translated into a declared CMB comparison coordinate and tested without allowing the branch parameters to drift.

The present work focuses on CMB comparison-map closure as a field-coordinate translation problem. The temperature map is not treated merely as a heat-map image: it is the inverse-harmonic representation of the angular structure carried by the translated comparison coordinate. The target is not a full Planck posterior and not a replacement for the official Planck likelihood. The target is narrower and falsifiable:

$$\begin{array}{ccc}
 \text{locked CPTG CMB envelope} & & \\
 \downarrow & & \\
 \text{real CMB map coordinate} & & (1) \\
 \downarrow & & \\
 \text{same residual metric as Planck} & &
 \end{array}$$

If the CPTG envelope were merely arbitrary, it should not remain adjacent to the Planck baseline after component-map changes, split-map changes, smoothing controls, sky-selection controls, and null-envelope tests.

2 Claim level

The appropriate claim level is a CMB field-coordinate closure result expressed through comparison maps. The paper tests whether a locked geometric row translates consistently into the observed CMB harmonic field coordinate and remains closed after reconstruction into real map space. It does not claim final Planck lowE, lensing, TTTEEE posterior, or combined-stack validation.

The primary claim is:

$$\text{locked CPTG geometry closes the observed CMB field coordinate} \quad (2)$$

under a locked geometric-pi branch and a declared phase-locked CMB translation protocol.

The observed phase scaffold is part of the observational field coordinate into which the locked CPTG envelope is translated. The meaningful test is whether the fixed CPTG angular-power structure can occupy that measured harmonic coordinate, reconstruct the temperature field with a residual nearly as small as the Planck baseline, and survive real-map and finite null-envelope controls.

The paper also keeps the spectrum-space and likelihood-grade programs outside the primary evidence chain. Binned TT/TE/EE closure, Planck lowE, Planck lensing, and full TTTEE nuisance-posterior validation are important follow-on tests, but they are different statistical objects from the real-map comparison closure tested here. They are therefore disclosed as future work rather than merged into the primary result.

3 Reader roadmap and evidence hierarchy

The paper is organized as an audit trail rather than as a single fitted number. The evidence hierarchy is shown in Table 1. The visual and component-map sections carry the primary comparison-map closure result. The smoothing, mask, and control-scatter sections test attribution of the remaining residual. The null-envelope section tests whether phase locking alone would make the tested generic null envelopes look successful. Spectrum-space TT/TE/EE work is deliberately excluded from the primary chain and is reserved for a separate paper.

Layer	Sections	Question answered	Claim status
Protocol and data	Data files, locked row, method	What map products, fields, masks, and equations define the test?	Setup
Primary map closure	Visual map, component maps, split maps, WMAP bands	Does the locked CPTG envelope remain adjacent to the Planck envelope in real CMB map coordinates?	Primary evidence
Attribution controls	Smoothing, mask, sky fraction, control scatter	Is the remaining Planck-over-CPTG edge larger than map-transfer and sky-selection control variation?	Diagnostic control
Discrimination controls	Null envelopes and perturbation basin	Is the near-degeneracy automatic under phase locking, or does the CPTG envelope carry real structure?	Diagnostic control
Future work	Spectrum-space and likelihood-grade extensions	What is not claimed in this map-closure paper?	Not primary here

Table 1: Reader roadmap and evidence hierarchy. The paper’s primary claim is map-space comparison closure. Attribution and null-envelope rows strengthen that claim but do not convert it into a full Planck likelihood or spectrum-space validation.

4 Pi-Based vs. Planck Likelihood Comparison

The Planck method is an observationally optimized cosmological-parameter and nuisance-likelihood framework. Its baseline spectrum is selected inside a likelihood system that contains cosmological parameters, foreground amplitudes, beam terms, calibration-like terms, and other observational nuisance quantities.

The CPTG pi-based comparison is different. The CPTG branch is fixed before the data product is evaluated:

$$\Theta_{\pi} = \{p_C, p_{ac}, G_T, D_c, A_s, T_{\tau}, H_0, r_d, N_{\text{eff}}, S_8\}_{\text{locked}}. \quad (3)$$

Here Θ_π denotes the upstream locked branch and gate registry; the registry symbols and their parent-level definitions are given in Ref. [8]. In particular, D_c , T_τ , r_d , S_8 , and the upstream amplitude conventions are parent-registry quantities rather than additional map-space fit parameters. The downstream CMB comparison coordinates used by the map-envelope calculation are listed separately below, together with the fixed CAMB solver assumptions explicitly set by the supplied scripts. In the downstream table, A_s means the numerical scalar-amplitude value actually passed to CAMB by the retained scripts. The upstream registry and downstream CMB coordinate should not be read as interchangeable parameter lists.

The observational sector then determines the comparison coordinate:

$$\Theta_\pi \longrightarrow m_s(\Theta_\pi, \nu_s), \quad (4)$$

where m_s is the model vector in the observational coordinate and ν_s denotes survey or pipeline nuisance quantities. These nuisance quantities may describe calibration, beam, foreground, mask, or component-separation behavior. They are not new CPTG branch parameters.

In a fitted tuned comparison, the theory-side parameters may move to improve one data sector:

$$\Theta_s^* = \arg \min_{\Theta_s} \chi_s^2. \quad (5)$$

That is not the protocol used here. The CPTG row is locked, transported into the CMB comparison-map coordinate, and then compared.

Table 2: Difference between a scalable pi-based CMB comparison and a Planck-style optimized comparison.

Feature	Pi-based CPTG comparison	Planck-style optimized comparison
Theory branch	Locked before CMB map evaluation	Cosmological parameters are optimized within a likelihood framework
Comparison object	Transported comparison-coordinate envelope	Best-fit or baseline CMB spectrum and nuisance profile
Allowed movable quantities	Observational nuisance or control quantities only	Cosmological and nuisance quantities may be optimized according to the likelihood protocol
Falsifiability	Stronger as a branch-coherence test because CPTG parameters do not move locally	Stronger as a precision cosmological inference framework
Interpretation of success	One locked geometric branch survives projection into real CMB map coordinates	A parameterized model fits the CMB likelihood under its intended machinery

5 Data files and map products

The analysis uses real CMB temperature maps and derived comparison products. The original Planck and WMAP FITS maps are not embedded in the manuscript package because several are large survey data products. The Planck component and split products should be obtained from the Planck public data archive or the equivalent Planck release

mirror, and the WMAP ILC products should be obtained from the NASA LAMBDA archive [4, 5]. The reproducibility package records the exact expected file names, fields, masks, scripts, output tables, and figures. A local reproduction requires placing the listed survey files in the working directory before running the command ledger in Appendix A.

All primary map-space rows are generated from HEALPix temperature maps. Each map is read from its declared temperature field, converted to microkelvin if necessary, downgraded to the declared working resolution, masked by the declared temperature mask, cleaned by monopole and dipole removal on the valid sky, and then projected into a phase-locked comparison coordinate. The default working resolution is

$$N_{\text{side}} = 256, \quad (6)$$

with

$$\ell_{\text{max}} = 3N_{\text{side}} - 1 = 767. \quad (7)$$

5.1 Primary Planck component-map files

The component-map test uses Planck temperature component-separation products. The frozen evidence rows use the files listed in Table 3. The full-map rows use `I_STOKES`; the inpainted-control rows use `I_STOKES_INP` when present. The temperature mask is `TMASK`, with `TMASKINP` used for inpainted-control rows when present. The supplied scripts retain an R3.01 SEVEM fallback for local portability, but the retained evidence metadata identifies R3.00 as the SEVEM product actually used.

Component	Expected local file name	Temperature field	Mask field
SMICA	<code>COM_CMB_IQU-smica_2048_R3.00_full.fits</code>	<code>I_STOKES</code> or <code>I_STOKES_INP</code>	<code>TMASK</code> or <code>TMASKINP</code>
SMICA-noSZ	<code>COM_CMB_IQU-smica-nosz_2048_R3.00_full.fits</code>	<code>I_STOKES</code> or <code>I_STOKES_INP</code>	map mask or SMICA mask fallback
NILC	<code>COM_CMB_IQU-nilc_2048_R3.00_full.fits</code>	<code>I_STOKES</code> or <code>I_STOKES_INP</code>	map mask or SMICA mask fallback
SEVEM	<code>COM_CMB_IQU-sevem_2048_R3.00_full.fits</code>	<code>I_STOKES</code> or <code>I_STOKES_INP</code>	map mask or SMICA mask fallback
Commander	<code>COM_CMB_IQU-commander_2048_R3.00_full.fits</code>	<code>I_STOKES</code> or <code>I_STOKES_INP</code>	map mask or SMICA mask fallback

Table 3: Planck component-map files used for the component robustness and inpainted-field controls. File names are local working names expected by the scripts. The original survey files should be obtained from the Planck Legacy Archive or equivalent Planck data release mirrors.

5.2 Planck split-map and WMAP files

The split-map test checks whether the CPTG-minus-Planck comparison gap is stable across internal Planck SMICA data splits. The WMAP row is a cross-mission low-multipole

control, used only in low- ℓ band-limited comparisons because the WMAP ILC map has a different transfer and smoothing history from the Planck component maps.

Layer	Expected local file name	Role
SMICA full	COM_CMB_IQU-smica_2048_R3.00_full.fits	Full-map reference in the split stability ladder
SMICA HM1	COM_CMB_IQU-smica_2048_R3.00_hm1.fits	Half-mission 1 internal stability row
SMICA HM2	COM_CMB_IQU-smica_2048_R3.00_hm2.fits	Half-mission 2 internal stability row
SMICA OE1	COM_CMB_IQU-smica_2048_R3.00_oe1.fits	Odd-ring internal stability row
SMICA OE2	COM_CMB_IQU-smica_2048_R3.00_oe2.fits	Even-ring internal stability row
WMAP ILC	wmap_ilc_9yr_v5.fits	Cross-mission low- ℓ phase-locked band comparison
WMAP mask, optional	wmap_temperature_analysis_mask_r9_9yr_v5.fits	Optional WMAP temperature-analysis mask

Table 4: Planck split-map and WMAP files used for internal data-split stability and cross-mission low- ℓ controls.

5.3 Map preprocessing protocol

Each HEALPix input map is processed with the same order of operations. The finite temperature map is written as T_i , where i indexes HEALPix pixels. The mask fraction after downgrade is M_i . The valid sky is defined by

$$W_i = \begin{cases} 1, & T_i \text{ finite and } M_i \geq M_{\text{cut}}, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

where the default temperature-mask threshold is

$$M_{\text{cut}} = 0.99. \quad (9)$$

For the mask ladder, M_{cut} is varied over

$$M_{\text{cut}} \in \{0.50, 0.70, 0.80, 0.90, 0.95, 0.99\}. \quad (10)$$

The Galactic latitude ladder applies additional cuts

$$|b| > 10^\circ, \quad |b| > 20^\circ, \quad |b| > 30^\circ, \quad |b| > 40^\circ. \quad (11)$$

The monopole and dipole are removed only on the valid sky. This prevents the invalid mask region from entering the fitted valid-sky field. For the harmonic translation step, invalid pixels are then filled with zero before the HEALPix harmonic transform. The resulting $a_{\ell m}^{\text{obs}}$ and C_ℓ^{obs} therefore define a masked pseudo-harmonic comparison scaffold rather than a beam-deconvolved full-sky realization. In the smoothing ladder, the final script uses mask-normalized smoothing,

$$T_{\text{smooth}}(\hat{n}) = \frac{\text{smooth}[T(\hat{n})W(\hat{n})]}{\text{smooth}[W(\hat{n})]}, \quad (12)$$

which prevents masked-sky non-finite pixels from poisoning the smoothed map.

5.4 Analysis scripts and output products

The evidence chain is generated by a set of small scripts rather than by a single monolithic analysis. Table 5 lists the scripts, their output products, and the manuscript role of each product. The package delivered with this paper includes the scripts and CSV outputs used to populate the tables and figures.

Script	Main output tables or figures	Manuscript role
<code>cptg_cmb_visual_compare_v2.py</code>	<code>cmb_vs_cptg_side_by_side_fitted.png</code> , <code>comparison_summary.txt</code>	Full visual CMB versus fitted CPTG comparison
<code>cptg_planck_component_robustness_v4.py</code>	<code>component_cptg_vs_planck_delta.csv</code> , component plots	Full-band Planck component robustness
<code>cptg_planck_splitmap_stability.py</code>	<code>splitmap_cptg_vs_planck_delta.csv</code> , split stability summary	Internal Planck splitmap stability
<code>cptg_cmb_smoothing_ladder_v2.py</code>	<code>smoothing_ladder_delta.csv</code> , <code>gap_by_smoother.png</code>	Smoothing and scale-sensitivity control
<code>cptg_cmb_mask_ladder.py</code>	<code>mask_ladder_delta.csv</code> , <code>gap_by_galactic_cut.png</code>	Sky-fraction and mask-boundary attribution
<code>cptg_wmap_lowell_bandtest.py</code>	<code>wmap_band_cptg_vs_planck_delta.csv</code>	Cross-mission low- ℓ comparison
<code>cptg_cmb_control_scatter_audit.py</code>	<code>control_scatter_by_run.csv</code> , <code>attribution_table.csv</code>	Residual-attribution audit across all controls
<code>cptg_cmb_null_envelope_controls.py</code>	<code>null_envelope_summary_by_model.csv</code> , <code>cptg_planck_vs_nulls.png</code>	Test whether phase locking alone explains the result

Table 5: Script and output ledger for the CMB comparison-map closure evidence chain. Each script writes machine-readable CSV tables used by the paper, so the numerical rows can be audited independently from the manuscript text.

5.5 Reproducibility boundary

The paper package is designed to reproduce the analysis once the external survey maps are available locally. The package includes the TeX source, figures, CSV outputs, scripts, and run notes. It does not include the original Planck or WMAP FITS files. Therefore the reproducibility claim is:

$$\begin{array}{c}
 \text{scripts} + \text{CSV} + \text{figures} + \text{commands} \quad \text{reproducible} \\
 \xrightarrow{\text{external survey maps}} \\
 \text{full rerun reproducible.}
 \end{array} \tag{13}$$

This is the correct boundary for a manuscript built from public survey data products whose raw map files are too large to embed in the article package.

6 Locked CMB comparison rows

The locked geometric-pi branch supplies the following CMB comparison coordinates:

$$p_C = \pi, \quad (14)$$

$$p_{ac} = 3 - \frac{\pi}{100} = 2.968584073464, \quad (15)$$

$$G_T = \frac{p_{ac}}{p_C} = 0.9449296585513721. \quad (16)$$

The main CMB comparison coordinates and fixed solver assumptions explicitly set by the supplied CAMB scripts are:

Quantity or setting	Value used	Role
H_0	67.4777967351 km s ⁻¹ Mpc ⁻¹	CPTG comparison coordinate
$\Omega_b h^2$	0.022527857494	CPTG comparison coordinate
$\Omega_c h^2$	0.117685841620526	CPTG comparison coordinate
n_s	0.968584073464	CPTG comparison coordinate
N_{eff}	2.968584073464	CPTG comparison coordinate
A_s	$2.136283004441 \times 10^{-9}$	CPTG comparison coordinate
τ	0.058930875934	CPTG comparison coordinate
A_{lens}	1	Fixed comparison setting
Ω_k	0	Fixed CAMB solver assumption
$\sum m_\nu$	0.06 eV	Fixed CAMB solver assumption
Helium handling	CAMB internal/default handling	Not explicitly overridden in the frozen map scripts

Table 6: CPTG CMB comparison coordinates and fixed CAMB assumptions explicitly used by the map-envelope scripts. Solver assumptions are listed separately from CPTG comparison coordinates so they are not mistaken for native branch quantities.

The parent pi-branch comparison registry separately records $Y_{\text{He,CMB}} = 0.2485$ as a frozen CMB comparison input [8]. The frozen map scripts in the present evidence chain do not explicitly pass `YHe` to CAMB, so this paper does not relabel that parent helium value as an explicitly applied input of the retained map-space runs. Other CAMB defaults not overridden by the scripts remain solver-side implementation choices rather than CPTG branch parameters.

The retained scripts generate the Planck benchmark envelope from the following fixed control row:

Quantity or setting	Planck control value	Role
H_0	$67.36 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Planck benchmark coordinate
$\Omega_b h^2$	0.02237	Planck benchmark coordinate
$\Omega_c h^2$	0.1200	Planck benchmark coordinate
n_s	0.9649	Planck benchmark coordinate
N_{eff}	3.046	Planck benchmark coordinate
A_s	2.100549×10^{-9}	Planck benchmark coordinate
τ	0.0544	Planck benchmark coordinate
A_{lens}	1	Fixed comparison setting
Ω_k	0	Fixed CAMB solver assumption
$\sum m_\nu$	0.06 eV	Fixed CAMB solver assumption

Table 7: Fixed Planck control row used to generate the benchmark CMB comparison envelope in the retained scripts.

CAMB is used here as the numerical coordinate-construction engine that maps each declared comparison row into a CMB angular-power envelope. Its background cosmological interpretation is not adopted as CPTG native dynamics: the CPTG theory input remains the locked geometric row, and the CAMB output used by this paper is the translated CMB comparison envelope.

7 Map-space method

For an observed CMB temperature map $T_{\text{obs}}(\hat{n})$, the analysis constructs masked pseudo-harmonic coefficients $a_{\ell m}^{\text{obs}}$ after downgrade, valid-sky monopole/dipole removal, and zero filling of invalid pixels. The corresponding pseudo-spectrum scaffold is C_ℓ^{obs} . These quantities define the comparison coordinate used here; they are not asserted to be a full-sky beam-deconvolved realization. For a model envelope C_ℓ^M , the phase-locked comparison coefficients are:

$$a_{\ell m}^M = a_{\ell m}^{\text{obs}} \left(\frac{C_\ell^M}{C_\ell^{\text{obs}}} \right)^{1/2}, \quad \ell \geq 2. \quad (17)$$

Because C_ℓ^{obs} is computed from the same masked pseudo-harmonic scaffold, the phase-locked construction satisfies within this comparison coordinate, for $\ell \geq 2$ wherever $C_\ell^{\text{obs}} > 0$,

$$\frac{1}{2\ell + 1} \sum_m |a_{\ell m}^M|^2 = C_\ell^M. \quad (18)$$

Thus the operation is exactly a spectral-envelope transplant onto the masked pseudo-harmonic phase coordinate defined by the declared preprocessing. The identity is a property of the CMB translator coordinate: the observed phase scaffold supplies the observational harmonic coordinate, while the locked model supplies the angular-power envelope carried into that coordinate. This is the mathematical core of the closure test. The CPTG construction is not being judged by whether a rendered heat map looks similar by eye; its fixed angular-power structure is imposed coherently across the retained

multipoles and their spherical-harmonic modes before the field is reconstructed in real space.

The resulting comparison map is:

$$T_M(\hat{n}) = \sum_{\ell m} a_{\ell m}^M Y_{\ell m}(\hat{n}). \quad (19)$$

The map-space envelope comparison then fits only an amplitude and offset:

$$T_{M,\text{fit}}(\hat{n}) = A_M \tilde{T}_M(\hat{n}) + B_M, \quad (20)$$

where $\tilde{T}_M$ is the normalized comparison-template shape on the valid sky. The residual fraction is:

$$R_M = \frac{\text{RMS}[T_{\text{obs}}(\hat{n}) - T_{M,\text{fit}}(\hat{n})]}{\text{RMS}[T_{\text{obs}}(\hat{n})]}. \quad (21)$$

With the declared monopole removal, the valid-sky observed mean is zero to numerical precision. Since $\tilde{T}_M$ is centered and normalized, the amplitude-plus-offset least-squares fit also gives

$$A_M = \rho_M \sigma_{\text{obs}}, \quad R_M = \sqrt{1 - \rho_M^2}, \quad (22)$$

where ρ_M is the valid-sky correlation coefficient. The reported correlation and fitted residual fraction are therefore equivalent summaries of the same fitted shape agreement rather than independent evidence statistics.

The fitted metric also projects out one global envelope-amplitude degree of freedom. For any constant $q > 0$,

$$C_\ell^M \rightarrow q C_\ell^M \implies T_M \rightarrow \sqrt{q} T_M \implies \tilde{T}_M \text{ unchanged}, \quad (23)$$

and therefore

$$R_M[q C_\ell^M] = R_M[C_\ell^M], \quad \rho_M[q C_\ell^M] = \rho_M[C_\ell^M]. \quad (24)$$

The primary fitted comparison is consequently a normalized envelope-shape test modulo one global amplitude projection. Absolute map-amplitude information remains visible in the separately reported raw visual row; it is not an independent degree of freedom in the headline fitted residual metric.

The comparison gap is:

$$\Delta R = R_{\text{CPTG}} - R_{\text{Planck}}. \quad (25)$$

Positive ΔR means the Planck baseline has the smaller residual under the selected metric.

8 Visual comparison map

The visual comparison is the most direct representation of the closure result, but the picture is not the test by itself. The observed map is the Planck SMICA full-mission temperature field after the declared preprocessing protocol, and its harmonic scaffold defines the observational field coordinate. The locked CPTG angular-power envelope is translated through that coordinate and then reconstructed as a temperature map. The Planck benchmark is treated identically. The question is therefore whether the fixed CPTG geometry closes the measured CMB field coordinate across its retained angular structure, with the map serving as the real-space readout of that translation.

For a cleaned observed map $T_{\text{obs}}(\hat{n})$, the phase scaffold is obtained from the observed harmonic coefficients. The raw CPTG comparison map is

$$T_{\text{CPTG}}^{\text{raw}}(\hat{n}) = \mathcal{Y}^{-1} \left[a_{\ell m}^{\text{obs}} \left(\frac{C_{\ell}^{\text{CPTG}}}{C_{\ell}^{\text{obs}}} \right)^{1/2} \right]. \quad (26)$$

The fitted visual comparison row then applies the same amplitude-plus-offset projection used in the map-space envelope comparisons:

$$T_{\text{CPTG}}^{\text{fit}}(\hat{n}) = A Z [T_{\text{CPTG}}^{\text{raw}}(\hat{n})] + B, \quad (27)$$

where $Z[\cdot]$ denotes valid-sky mean subtraction and unit-RMS normalization before the fitted amplitude and offset are applied. The residual image is

$$\epsilon_{\text{CPTG}}^{\text{fit}}(\hat{n}) = T_{\text{obs}}(\hat{n}) - T_{\text{CPTG}}^{\text{fit}}(\hat{n}). \quad (28)$$

The residual field is the meaningful visual diagnostic because the observed and CPTG panels intentionally share phase support.

Item	Value
CMB source map	COM_CMB_IQU-smica_2048_R3.00_full.fits
Temperature field	I_STOKES
Mask field	TMASK
Unit conversion	K_CMB to μK
Analysis resolution	$N_{\text{side}} = 256$
Mask threshold	0.99
Valid sky fraction	0.836431
Monopole/dipole treatment	Removed on the valid sky after downgrade and masking

Table 8: Source-map and preprocessing details for the SMICA visual comparison.

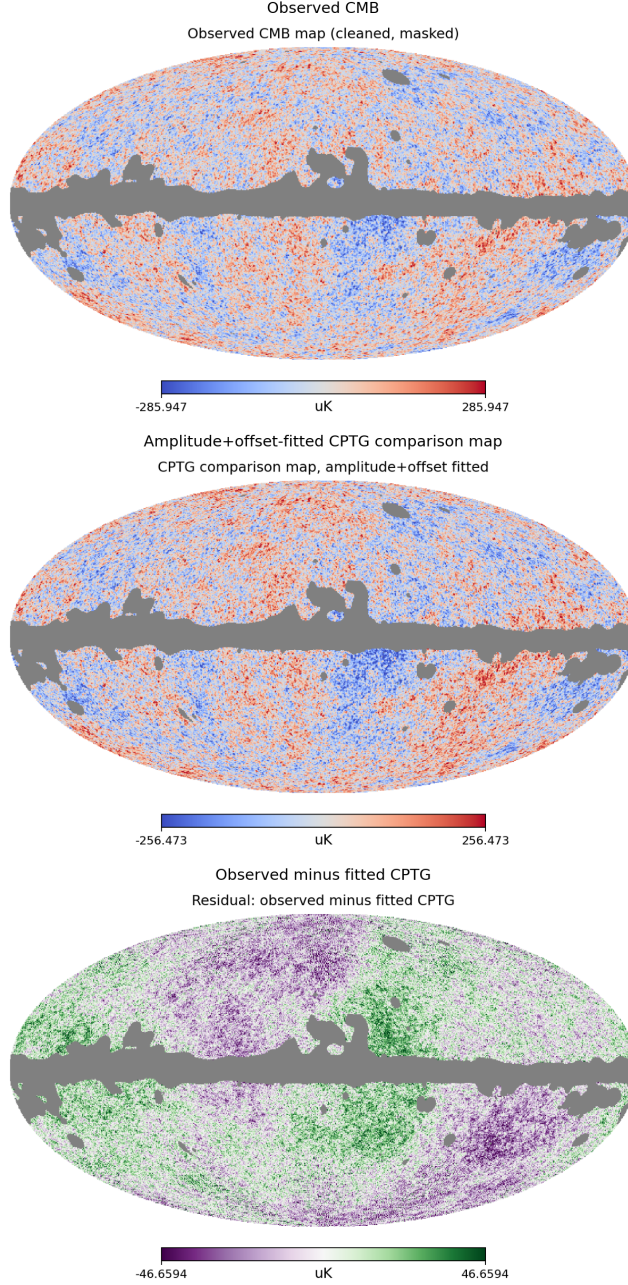


Figure 1: Observed SMICA CMB map, amplitude-plus-offset fitted CPTG comparison map, and observed-minus-fitted-CPTG residual. The first two panels share the same observed phase scaffold. After the fitted global amplitude and offset are projected out, the bottom panel shows the remaining normalized envelope-shape and reconstruction mismatch in the same map coordinate.

The fitted visual map is the comparison-coordinate image used for the primary map-space reading. The observed map has valid-sky RMS $100.509653 \mu\text{K}$. The fitted CPTG comparison map has RMS $98.844631 \mu\text{K}$, fitted amplitude $98.844631 \mu\text{K}$ per template standard deviation, and fitted offset consistent with zero. The observed-minus-fitted-CPTG residual RMS is $18.218928 \mu\text{K}$, with valid-sky correlation 0.983434 .

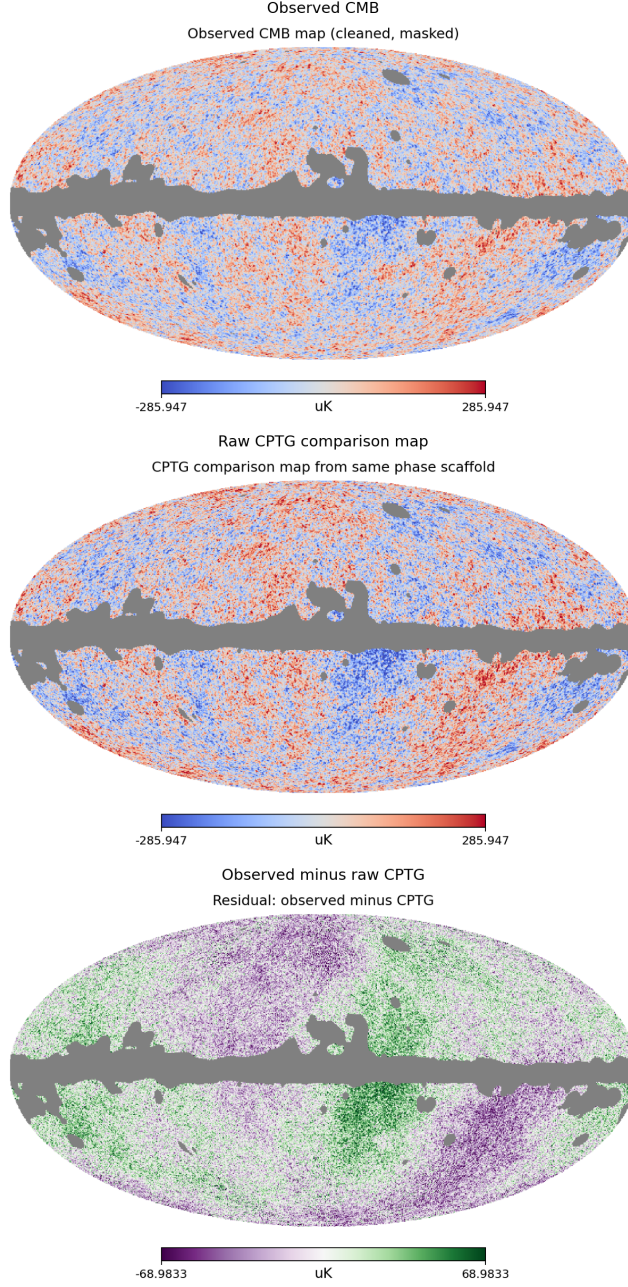


Figure 2: Observed SMICA CMB map, raw CPTG comparison map, and observed-minus-raw-CPTG residual. This row is useful because it shows the unscaled envelope projection. Its residual is larger than the fitted visual row because no amplitude-plus-offset projection has been applied.

Quantity	Observed	Raw CPTG	Fitted CPTG	Interpretation
RMS [μK]	100.509653	118.452082	98.844631	Temperature-amplitude scale
Residual RMS [μK]	–	26.784740	18.218928	Fitted row is metric-consistent
Observed-CPTG correlation	–	0.983434	0.983434	Amplitude fitting preserves correlation
Fitted amplitude [μK]	–	–	98.844631	Valid-sky template scale
Fitted offset [μK]	–	–	≈ 0	No meaningful monopole offset remains

Table 9: Raw and fitted visual comparison metrics for the SMICA full map.

The raw and fitted rows answer different questions. The raw row asks what the locked

CPTG envelope gives before map-amplitude projection. The fitted row asks what remains after the same valid-sky amplitude-plus-offset projection used by the envelope-competition metric. Since the residual RMS falls from $26.784740 \mu\text{K}$ to $18.218928 \mu\text{K}$, a substantial part of the raw difference is global amplitude-scale response. The fitted row then tests normalized envelope shape and reconstruction mismatch within the shared phase coordinate.

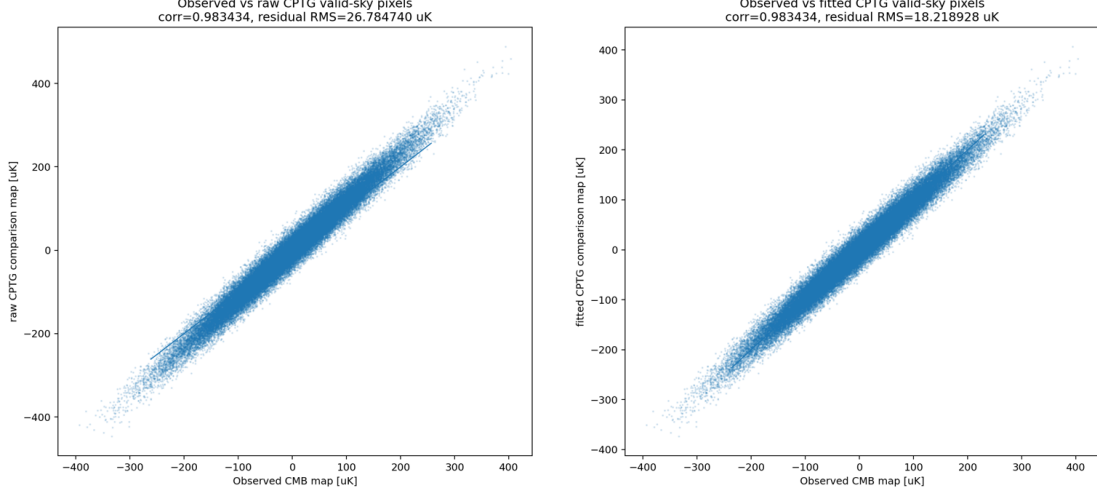


Figure 3: Valid-sky pixel scatter for the raw and fitted CPTG comparison maps. The tight diagonal structure shows that the comparison map preserves the observed hot/cold-spot organization. The fitted row reduces residual RMS; positive linear amplitude fitting preserves correlation.

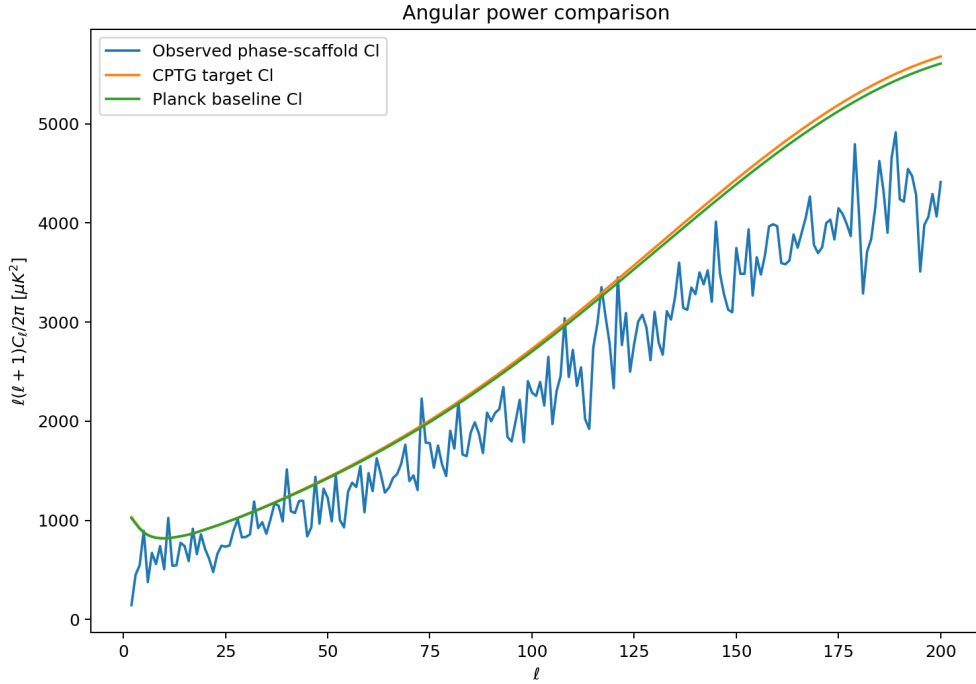


Figure 4: Angular-power comparison associated with the visual map run. The observed phase-scaffold spectrum, locked CPTG target envelope, and Planck baseline envelope are shown in the same $l(l+1)C_l/2\pi$ coordinate. The figure displays the angular-power quantities carried by the CMB coordinate translation.

The visual evidence therefore supports a precise claim: the locked CPTG comparison envelope, when carried through the real SMICA phase scaffold and the declared map-space translation, reconstructs the observed CMB temperature organization with correlation 0.983434 and fitted residual RMS 18.218928 μK . The locked geometric-pi construction therefore closes the observed map-space comparison coordinate at a level that is visually and metrically near-degenerate with the standard Planck comparison envelope.

9 Headline map-space results

The headline evidence chain is the map-space comparison chain. The same observed phase scaffold is used for the CPTG and Planck comparison maps, the same sky mask is applied, and the same amplitude-plus-offset residual metric is used. The result is not that CPTG wins every row. The result is that the locked CPTG geometric-pi comparison envelope remains adjacent to the Planck baseline envelope at a sub-microkelvin residual-gap scale across distinct real-map controls.

Let

$$R_M = \frac{\sqrt{\langle (T_{\text{obs}} - T_{M,\text{fit}})^2 \rangle_{\text{mask}}}}{\sqrt{\langle T_{\text{obs}}^2 \rangle_{\text{mask}}}} \quad (29)$$

be the fitted map-space residual fraction for model envelope M . The reported gap is

$$\Delta\sigma_{\text{res}} = \sqrt{\langle (T_{\text{obs}} - T_{\text{CPTG,fit}})^2 \rangle_{\text{mask}}} - \sqrt{\langle (T_{\text{obs}} - T_{\text{Planck,fit}})^2 \rangle_{\text{mask}}}. \quad (30)$$

Positive values mean the Planck baseline is ahead under this selected metric. A negative value means the CPTG comparison map is ahead for that row.

Run family	Rows	$\langle R_{\text{CPTG}} \rangle$	$\langle R_{\text{Planck}} \rangle$	Mean RMS gap [μK]	Sign status
Planck full-band non-inpainted	5	0.181485	0.181251	0.023518	Planck ahead
Planck full-band inpainted	5	0.182665	0.182403	0.026408	Planck ahead
Planck low- ℓ bands non-inpainted	20	0.220746	0.219838	0.044683	Planck ahead
Planck low- ℓ bands inpainted	20	0.227068	0.226190	0.044103	Planck ahead
WMAP low- ℓ bands	4	0.223921	0.223144	0.041417	Planck ahead
Planck SMICA full-band split maps	5	0.181408	0.181170	0.023891	Planck ahead

Table 10: Headline map-space comparison summary. This table is the primary evidence ledger for the present paper. TT/TE/EE spectrum-space diagnostics are intentionally excluded.

The magnitude of the gap is the important scale. In full-band Planck component maps and split maps, the mean residual gap is about 0.024 μK , while the observed map RMS is about 100 μK . This is a comparison between two harmonic-field translations, not a visual pixel-matching exercise. The result therefore places the locked CPTG envelope and the Planck baseline in nearly the same residual basin after both are carried through the same observed CMB field coordinate.

The observed phase scaffold defines the map coordinate and is shared by construction. The nontrivial point is that generic null envelopes translated through that same coordinate do not reproduce the same performance, while the locked CPTG envelope remains adjacent to the Planck baseline.

10 Planck component-map closure

The first real-map closure test asks whether the result depends on a single Planck component-separation product. The non-inpainted component set uses SMICA, SMICA-noSZ, NILC, SEVEM, and Commander where available. The inpainted set repeats the comparison using inpainted temperature fields when the map product supplies them. This is a component-separation robustness test, not a new fit.

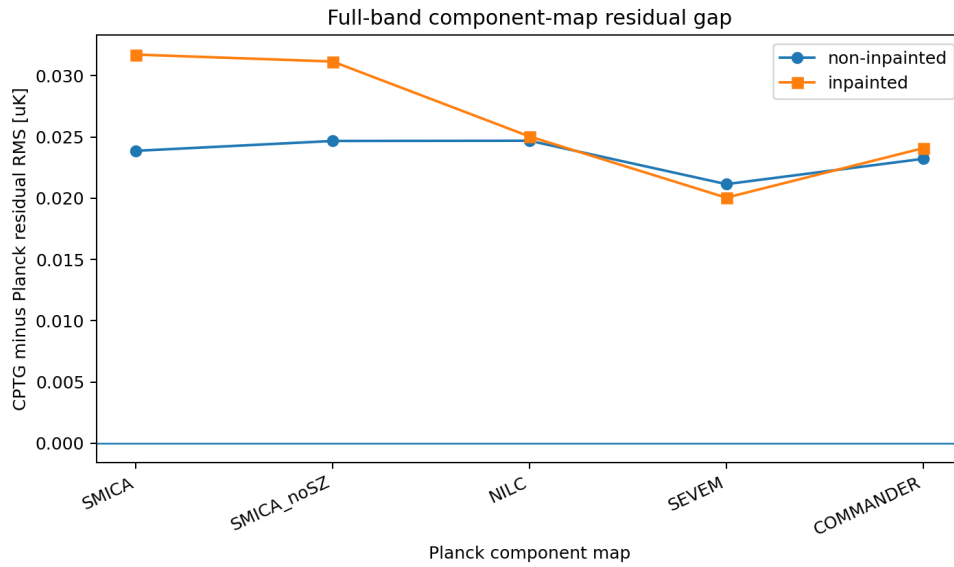


Figure 5: Full-band CPTG-minus-Planck residual RMS gap by Planck component map. Positive values mean the Planck baseline is ahead. The gap remains at the few $10^{-2} \mu\text{K}$ level across component products.

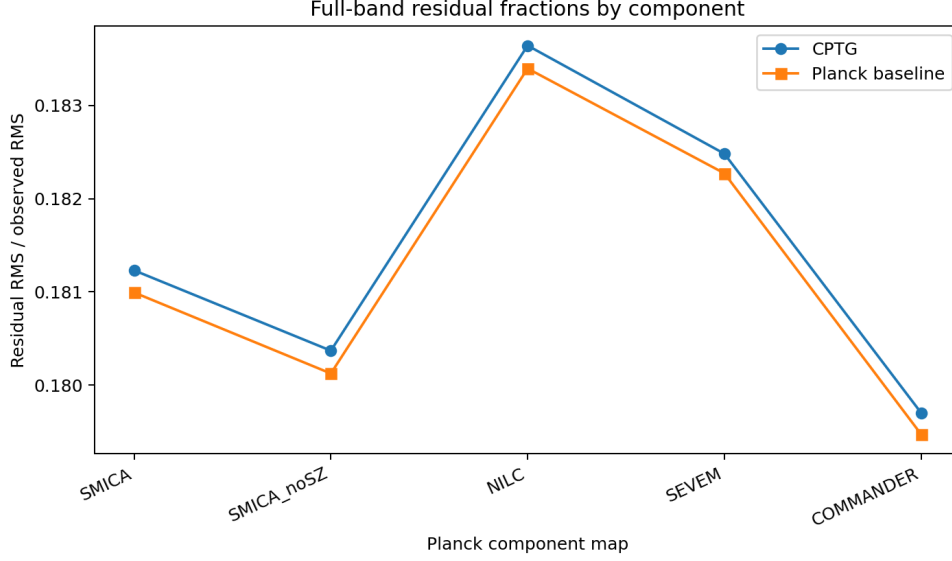


Figure 6: Full-band non-inpainted residual fractions for CPTG and the Planck baseline. The two curves are visually nearly degenerate at the residual-fraction scale.

Table 11: Full-band Planck component-map comparison. Positive $\Delta\sigma_{\text{res}}$ means the Planck baseline is ahead under the selected fitted map-space metric.

State	Component	Field	Mask	f_{sky}	R_{CPTG}	R_{Planck}	$\Delta\sigma_{\text{res}}$
Non-inpainted	SMICA	I_STOKES	TMASK	0.836	0.181230	0.180993	0.023866
Non-inpainted	SMICA-noSZ	I_STOKES	TMASK	0.836	0.180368	0.180123	0.024671
Non-inpainted	NILC	I_STOKES	TMASK	0.782	0.183644	0.183397	0.024689
Non-inpainted	SEVEM	I_STOKES	TMASK	0.838	0.182482	0.182272	0.021148
Non-inpainted	Commander	I_STOKES	TMASK	0.880	0.179700	0.179470	0.023214
Inpainted	SMICA	I_STOKES_INP	TMASKINP	0.983	0.183661	0.183347	0.031722
Inpainted	SMICA-noSZ	I_STOKES_INP	TMASKINP	0.983	0.186168	0.185861	0.031154
Inpainted	NILC	I_STOKES_INP	TMASK	0.782	0.183346	0.183096	0.025037
Inpainted	SEVEM	I_STOKES_INP	TMASKINP	0.961	0.181043	0.180845	0.020043
Inpainted	Commander	I_STOKES_INP	TMASK	0.880	0.179107	0.178868	0.024082

The component rows show three useful facts. First, the sign of the selected metric is stable: the Planck baseline is slightly ahead in each component row. Second, the numerical separation is extremely small. In the non-inpainted rows the RMS gap spans only 0.021148–0.024689 μK . Third, the same pattern survives inpainted fields, where the mean gap is 0.026408 μK .

This shows that the closure is not confined to the SMICA component-separation pipeline. It does not prove that component separation is irrelevant. Rather, it shows that component-product choice changes the absolute residual surface without dislodging CPTG from the same narrow Planck-like envelope basin.

Question	Result	Interpretation
Does the result depend only on SMICA?	No. Five Planck component products were tested.	The near-degeneracy is component-stable.
Does inpainting destroy the result?	No. The inpainted mean gap remains $0.026408 \mu\text{K}$.	Mask-fill treatment changes details but not the closure conclusion.
Does CPTG outperform Planck in this family?	No. Planck is slightly ahead in each component row.	The claim is near-Planck comparison-map closure, not a Planck defeat.
Is the gap large compared with the CMB map scale?	No. The gap is $\sim 0.02\text{--}0.03 \mu\text{K}$.	The separation is sub-microkelvin and attribution-limited.

Table 12: Component-map closure interpretation.

11 Band-limited Planck and WMAP closure

The band-limited tests restrict the comparison to declared low- ℓ ranges. These rows check whether the near-degeneracy is only a full-band artifact. They also provide a cross-mission layer through WMAP low- ℓ bands.

Dataset	Band	Rows	Mean ΔR	Mean RMS gap	RMS gap span
Planck non-inpainted	2 – –30	5	0.001199	0.051557	0.050458–0.052990
Planck non-inpainted	2 – –40	5	0.001066	0.050051	0.048896–0.051420
Planck non-inpainted	2 – –64	5	0.000819	0.043868	0.042653–0.045111
Planck non-inpainted	2 – –95	5	0.000548	0.033255	0.032319–0.034291
Planck inpainted	2 – –30	5	0.001129	0.049780	0.046673–0.052969
Planck inpainted	2 – –40	5	0.001017	0.048611	0.045791–0.051404
Planck inpainted	2 – –64	5	0.000800	0.043437	0.041350–0.045390
Planck inpainted	2 – –95	5	0.000565	0.034583	0.033300–0.037496
WMAP 9-year ILC	2 – –30	1	0.000873	0.040948	0.040948–0.040948
WMAP 9-year ILC	2 – –40	1	0.000801	0.040377	0.040377–0.040377
WMAP 9-year ILC	2 – –64	1	0.000700	0.039238	0.039238–0.039238
WMAP 9-year ILC	2 – –95	1	0.000736	0.045103	0.045103–0.045103

Table 13: Band-limited Planck and WMAP comparison. Planck rows are component-map means; WMAP rows are WMAP ILC band rows.

The low- ℓ band rows are not as numerically tight as the full-band residual-fraction rows because low- ℓ maps have smaller observed RMS and larger cosmic-variance sensitivity. Even so, the same qualitative result holds: the Planck baseline is slightly ahead, while the locked CPTG envelope remains close under the same phase-locked map construction. The WMAP rows are especially important because they repeat the comparison in a different mission product rather than another Planck component map.

The band-limited comparison is therefore a support layer for the map-space closure claim. It shows that the result is not created solely by high- ℓ structure or by one Planck component-separation pipeline.

12 Planck split-map stability

The split-map control repeats the comparison on the Planck SMICA full, half-mission, and odd/even products. This test asks whether the CPTG-minus-Planck residual gap is stable under internal Planck data splits rather than only under component-map changes.

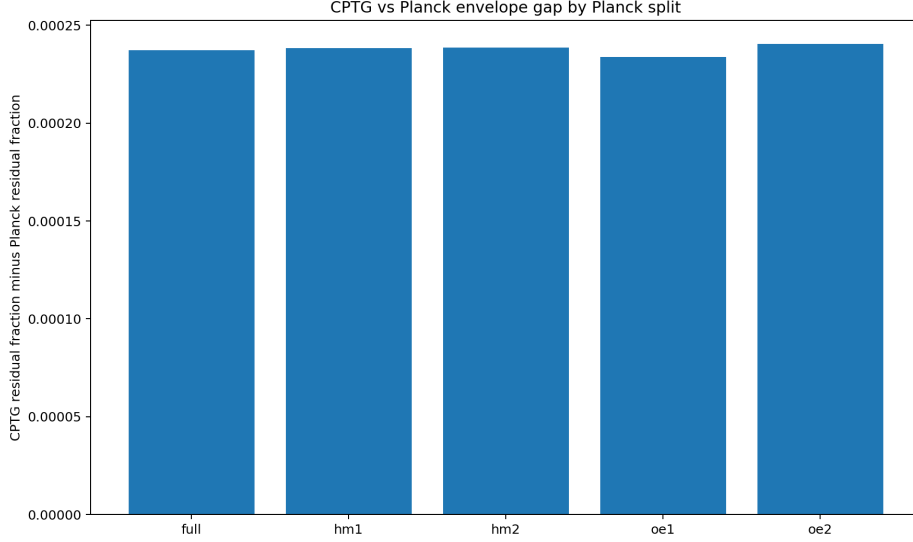


Figure 7: CPTG-minus-Planck residual gap across Planck SMICA split maps. The gap is extremely stable across full, half-mission, and odd/even split products.

Split	Observed RMS	R_{CPTG}	R_{Planck}	ΔR	RMS gap
Full	100.509653	0.181265	0.181028	0.000237	0.023851
HM1	100.566268	0.181383	0.181145	0.000238	0.023952
HM2	100.454668	0.181493	0.181255	0.000239	0.023972
OE1	100.478533	0.181369	0.181135	0.000234	0.023495
OE2	100.548897	0.181530	0.181289	0.000241	0.024184

Table 14: Full-band Planck SMICA split-map stability.

The mean full-band split-map RMS gap is

$$\langle \Delta \sigma_{\text{res}} \rangle_{\text{split}} = 0.023891 \mu\text{K}, \quad (31)$$

with a span across the full and split products of only

$$0.000690 \mu\text{K}. \quad (32)$$

This is the tightest stability result in the paper. The full-and-split product span is far smaller than the already small mean gap, so the relative CPTG-vs-Planck separation is not a random accident of the full SMICA map.

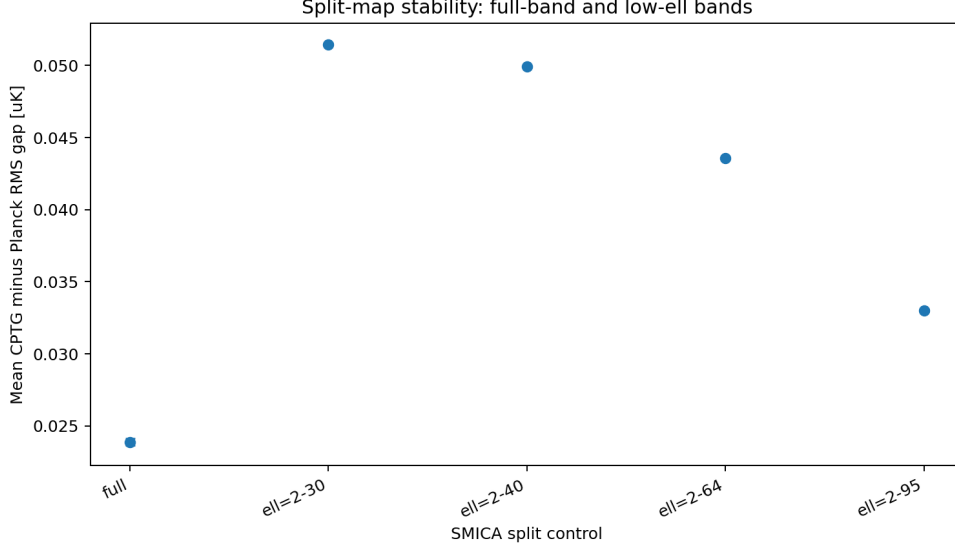


Figure 8: Split-map stability in the full band and in low- ℓ bands. Error bars show split-to-split standard deviation of the RMS gap.

Scope	Splits	Mean RMS gap	Std. RMS gap	Span RMS gap	Planck ahead all
Full band	5	0.023891	0.000253	0.000690	True
ell=2–30	5	0.051468	0.000054	0.000121	True
ell=2–40	5	0.049908	0.000073	0.000184	True
ell=2–64	5	0.043583	0.000090	0.000246	True
ell=2–95	5	0.033005	0.000102	0.000289	True

Table 15: Split-map stability summary for full-band and band-limited controls.

The band-limited split rows reinforce the same conclusion. The $\ell = 2\text{--}30$ and $\ell = 2\text{--}40$ split spans are only $0.000121\ \mu\text{K}$ and $0.000184\ \mu\text{K}$, respectively. Thus the low- ℓ comparison is also internally stable under Planck split-map changes.

13 Smoothing and scale sensitivity

The smoothing ladder asks whether the residual gap is stable under controlled changes in the effective beam and map-transfer scale. The test is not a new CPTG fit. For each smoothing scale, the observed map, the CPTG comparison map, and the Planck comparison map are all passed through the same mask-normalized Gaussian smoothing operation:

$$T_{\text{smooth}}(\hat{n}) = \frac{\text{smooth}[T(\hat{n})W(\hat{n})]}{\text{smooth}[W(\hat{n})]}. \quad (33)$$

The denominator prevents masked or non-finite sky pixels from contaminating the smoothed field. This control asks whether the CPTG-minus-Planck gap is scale invariant under the declared smoothing operator. It is not: the sign remains mostly Planck-ahead,

but the amplitude changes with smoothing scale. That observation establishes smoothing/scale sensitivity without, by itself, assigning the change uniquely to instrumental transfer or to envelope-shape physics.

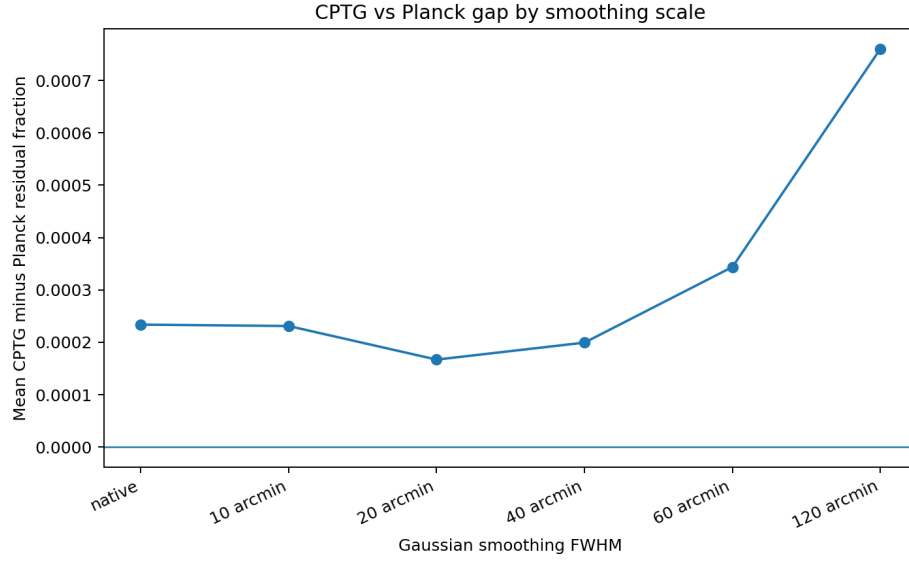


Figure 9: CPTG-minus-Planck gap as a function of Gaussian smoothing scale. The sign remains Planck-ahead, but the gap amplitude changes with smoothing scale.

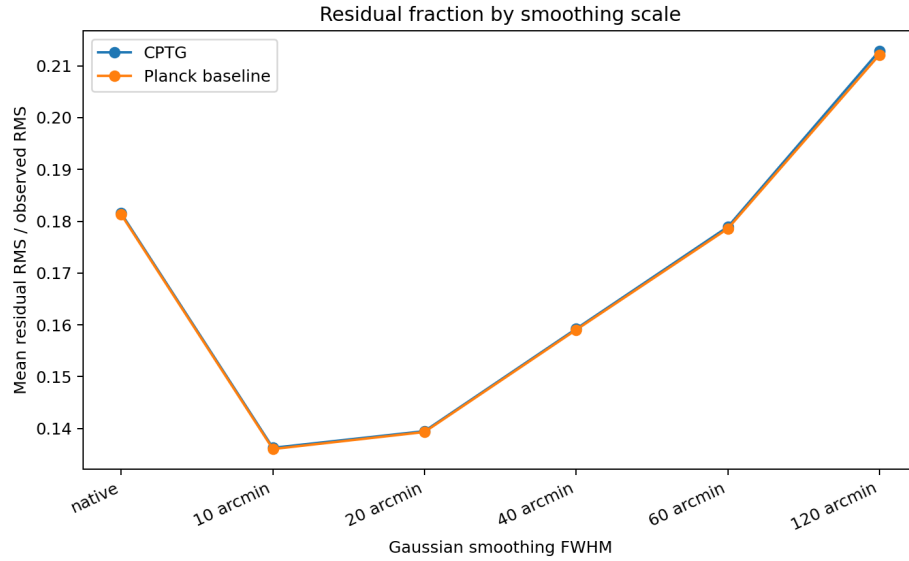


Figure 10: Mean CPTG and Planck residual fractions under the smoothing ladder. Both envelopes move together, while the small difference between them changes with smoothing scale.

Smoothing	Comp.	R_{CPTG}	R_{Planck}	ΔR	RMS gap	RMS span
native	5	0.181527	0.181293	+0.000234	0.023508	0.003565
10'	5	0.136266	0.136035	+0.000231	0.022297	0.004660
20'	5	0.139484	0.139317	+0.000167	0.014943	0.003758
40'	5	0.159239	0.159039	+0.000199	0.015169	0.002648
60'	5	0.178913	0.178569	+0.000344	0.022847	0.002046
120'	5	0.212889	0.212128	+0.000760	0.040205	0.002453

Table 16: Full-band smoothing ladder. Values are Planck component-map means. Positive gaps mean the Planck baseline has the smaller residual.

The full-band gap is smallest around the 20–40 arcmin smoothing scale and largest at 120 arcmin. The native row of the smoothing-control pipeline has mean gap $0.023508 \mu\text{K}$, while the separately tabulated primary component-map pipeline gives $0.023518 \mu\text{K}$; the $0.000010 \mu\text{K}$ implementation-level difference does not alter the differential closure result. The 120 arcmin smoothing mean is $0.040205 \mu\text{K}$. The ladder therefore shows that the residual gap is not scale invariant under the declared smoothing operator. Separating beam or map-transfer effects from scale-dependent envelope-shape effects requires explicit attribution beyond the smoothing ladder itself.

The band-limited smoothing ladder gives a more refined reading. The lowest bands remain nearly fixed, while the wider bands become more smoothing-sensitive.

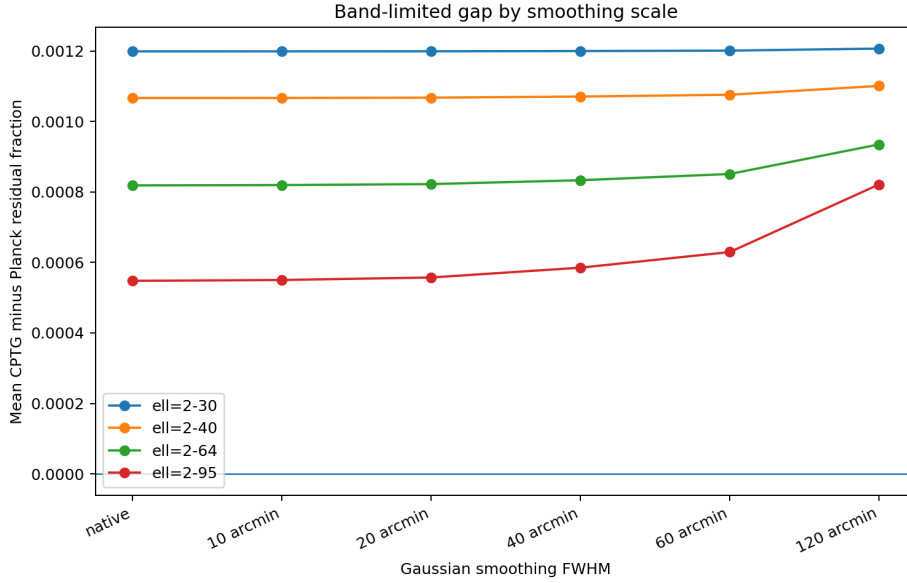


Figure 11: Band-limited CPTG-minus-Planck gap under Gaussian smoothing. The lowest bands are stable; wider bands show more smoothing response.

Band	Native RMS gap [μK]	120 arcmin RMS gap [μK]	Change [μK]
$\ell = 2 - -30$	0.051557	0.050815	-0.000741
$\ell = 2 - -40$	0.050051	0.049795	-0.000255
$\ell = 2 - -64$	0.043868	0.046069	+0.002201
$\ell = 2 - -95$	0.033255	0.042321	+0.009066

Table 17: Band-limited smoothing response. The table compares the native and 120 arcmin RMS gaps.

The smoothing result therefore supports the attribution-limited interpretation. The CPTG comparison envelope remains close to the Planck baseline across all smoothing scales, while the tiny Planck-over-CPTG edge changes under the declared smoothing operator. The proper claim is not that smoothing identifies a unique physical or instrumental cause. The proper claim is that attribution among beam/transfer effects, map construction, and scale-dependent envelope-shape differences remains unresolved.

14 Mask and sky-fraction attribution

The mask ladder tests whether the residual gap depends on sky selection. Two controls are separated. The first varies the Planck temperature-mask threshold. The second applies Galactic latitude cuts. These are not equivalent. The threshold ladder mostly changes the detailed mask boundary, while the Galactic latitude ladder changes the foreground and sky-region content of the valid-sky comparison.

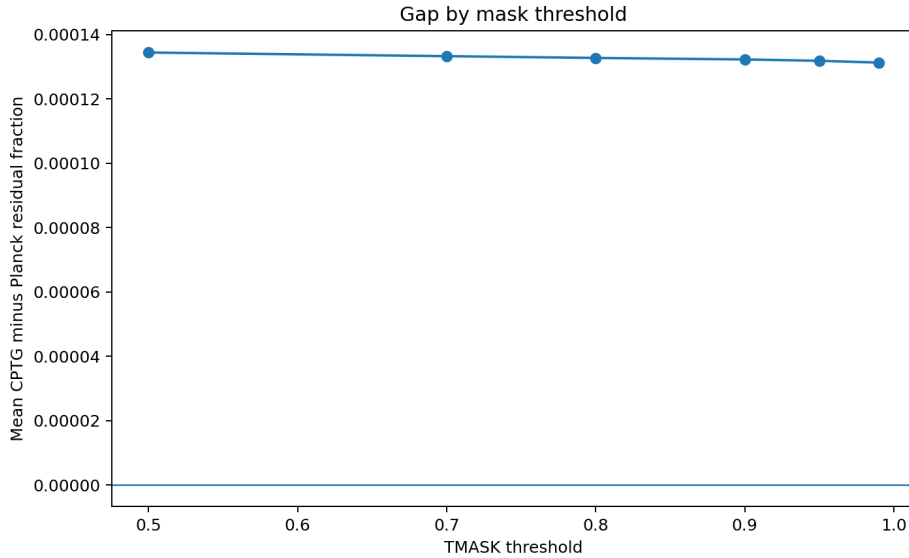


Figure 12: CPTG-minus-Planck gap by temperature-mask threshold. The threshold ladder is comparatively stable, indicating that the result is not a fragile single-threshold artifact.

Threshold	Rows	Mean f_{sky}	$\langle \Delta R \rangle$	Mean RMS gap [μK]	RMS span [μK]	CPTG-ahead rows
0.50	25	0.626048	+0.000134	0.013436	0.026019	1
0.70	25	0.625450	+0.000133	0.013322	0.025807	1
0.80	25	0.625103	+0.000133	0.013267	0.025702	1
0.90	25	0.624708	+0.000132	0.013220	0.025566	1
0.95	25	0.624452	+0.000132	0.013178	0.025462	1
0.99	25	0.624073	+0.000131	0.013120	0.025351	1

Table 18: Temperature-mask threshold ladder. Rows include all Galactic-cut states and all Planck component maps.

The threshold means run from $0.013436 \mu\text{K}$ at threshold 0.50 to $0.013120 \mu\text{K}$ at threshold 0.99. This is a very small threshold dependence. Therefore the map-space closure is not being produced by one special mask-threshold choice.

The Galactic latitude ladder has a different character. Removing the Galactic plane strongly reduces the Planck-over-CPTG edge.

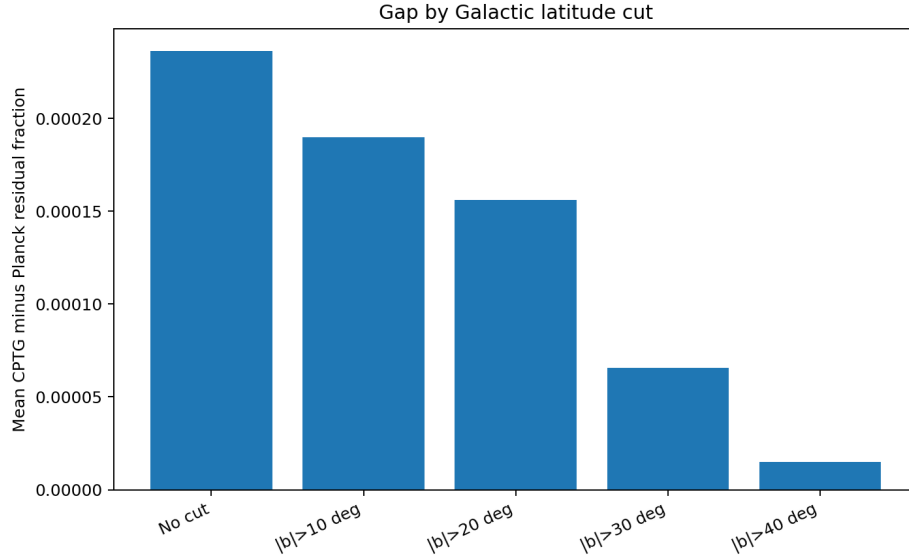


Figure 13: CPTG-minus-Planck residual gap by Galactic latitude cut. The all-sky Planck-over-CPTG edge shrinks strongly when the Galactic plane is removed.

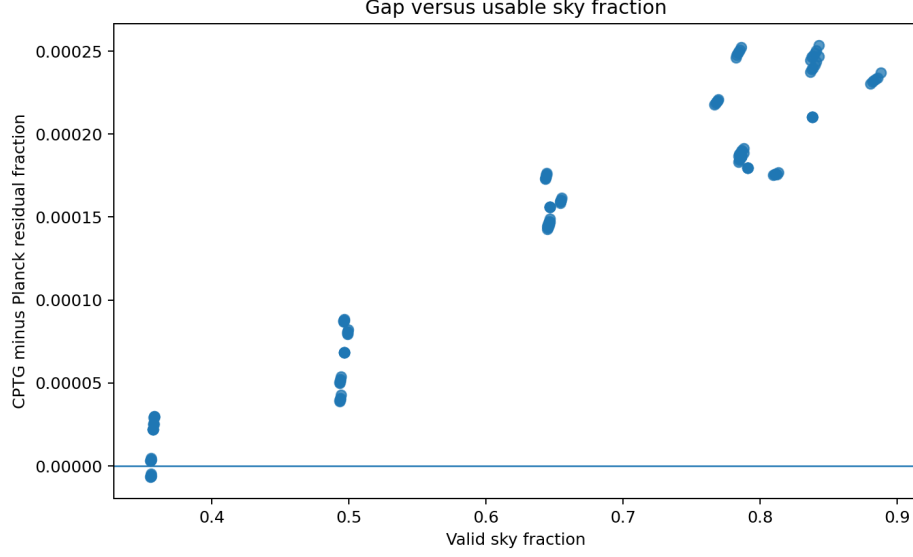


Figure 14: Gap versus usable sky fraction. The full-band gap approaches zero in the most restrictive high-latitude rows, and a small number of rows become CPTG-ahead.

Sky rule	Mean f_{sky}	$\langle \Delta R \rangle$	RMS gap [μK]	CPTG-ahead rows	Planck ahead all
No Galactic cut	0.836853	+0.000236	0.023788	0	True
$ b > 10^\circ$	0.788371	+0.000190	0.019013	0	True
$ b > 20^\circ$	0.647134	+0.000156	0.015545	0	True
$ b > 30^\circ$	0.495794	+0.000066	0.006477	0	True
$ b > 40^\circ$	0.356709	+0.000015	0.001463	6	False

Table 19: Galactic latitude sky-selection control. Values are Planck component means over mask thresholds.

The full-band mean RMS gap falls from $0.023788 \mu\text{K}$ with no Galactic cut to $0.001463 \mu\text{K}$ at $|b| > 40^\circ$. The high-latitude row also contains CPTG-ahead cases. This does not mean CPTG has been retuned. It means the residual edge is partly sky-selection dependent and should not be assigned uniquely to the theory branch.

The band-limited mask ladder reinforces the same conclusion. The band-limited gaps remain Planck-ahead in the mean, but the amplitude depends on the angular band and the sky cut.

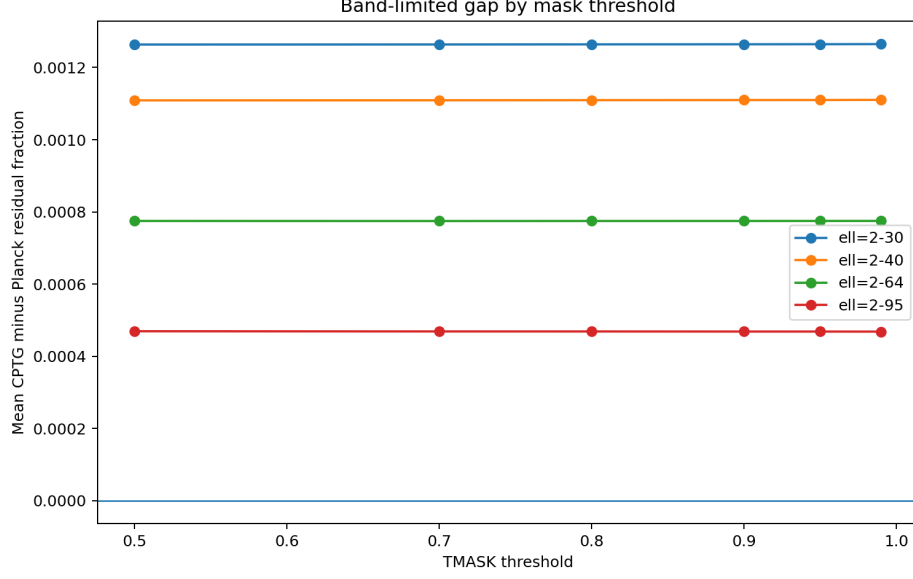


Figure 15: Band-limited mask ladder. The mask threshold itself is stable, while the gap depends more strongly on sky selection and angular band.

Band	Rows	$\langle \Delta R \rangle$	Mean RMS gap [μK]	RMS span [μK]	Planck-ahead rows
$\ell = 2 - -30$	150	+0.001264	0.051549	0.009356	100.0%
$\ell = 2 - -40$	150	+0.001110	0.049619	0.011246	100.0%
$\ell = 2 - -64$	150	+0.000775	0.040286	0.017973	100.0%
$\ell = 2 - -95$	150	+0.000469	0.027840	0.022150	100.0%

Table 20: Band-limited mask ladder summary. Rows include all Planck components, thresholds, and Galactic cuts.

The mask and sky-fraction controls strengthen the paper’s main claim boundary. CPTG remains extremely close to the Planck comparison envelope, but the remaining sub- μK separation is not a clean theory-only residual. It changes with sky selection and can nearly vanish under high-latitude cuts.

15 Control-scatter audit

The previous sections show that the CPTG and Planck envelopes are separated by a very small positive gap in most rows. The control-scatter audit asks a different question. It asks whether that gap is larger than the scatter introduced by map product, split map, smoothing, band selection, and sky selection. This is the key attribution test. A stable Planck-ahead sign can be measurable, but if the amplitude is smaller than the control span then the residual should not be assigned uniquely to CPTG.

The audit combines every map-space control row used in this paper into one comparison ledger. These rows are nested and correlated control evaluations rather than independent statistical trials; their means, standard deviations, spans, and sign fractions are descriptive

summaries of the sampled control grid. Each row uses the same residual-RMS gap,

$$\Delta\sigma_{\text{res}} = R_{\text{RMS}}^{\text{CPTG}} - R_{\text{RMS}}^{\text{Planck}}, \quad (34)$$

where positive values mean that the Planck baseline is ahead in the selected RMS metric. The important comparison is not only the mean value of $\Delta\sigma_{\text{res}}$, but also its standard deviation, span, and sign behavior across controls.

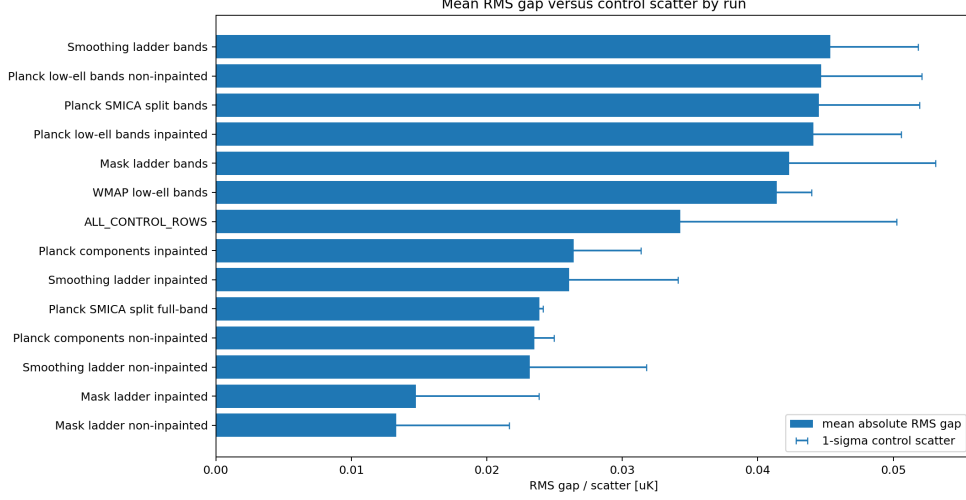


Figure 16: Control-scatter audit by run family. The horizontal bars show the mean absolute CPTG-minus-Planck RMS gap, while the error bars show control scatter. Runs for which the scatter or span is comparable to the mean gap are attribution-limited even when the sign is usually Planck-ahead.

Control run	Rows	Mean gap [μK]	Std. [μK]	Span [μK]	Planck-ahead	Sign flip
Planck components	5	0.023518	0.001460	0.003541	100.00%	no
Planck components, inpainted	5	0.026408	0.004964	0.011678	100.00%	no
Planck split maps, full band	5	0.023891	0.000253	0.000690	100.00%	no
Planck split maps, bands	20	0.044491	0.007448	0.018676	100.00%	no
Planck low- ℓ bands	20	0.044683	0.007430	0.020671	100.00%	no
Planck low- ℓ bands, inpainted	20	0.044103	0.006474	0.019669	100.00%	no
WMAP low- ℓ bands	4	0.041417	0.002558	0.005865	100.00%	no
Smoothing ladder	30	0.023161	0.008621	0.029317	100.00%	no
Smoothing ladder, inpainted	30	0.026067	0.008027	0.026374	100.00%	no
Smoothing ladder, bands	120	0.045343	0.006501	0.020671	100.00%	no
Mask ladder	150	0.013257	0.008333	0.026203	96.00%	yes
Mask ladder, inpainted	150	0.014752	0.009073	0.032826	98.67%	yes
Mask ladder, bands	600	0.042323	0.010802	0.042286	100.00%	no
All control rows	1159	0.034265	0.015964	0.056006	99.31%	yes

Table 21: Full control-scatter run ledger. Positive mean gaps indicate that the Planck baseline is ahead. Sign-flip rows contain at least one CPTG-ahead case. The rows are nested and correlated control evaluations, so the percentages and scatter values are descriptive rather than independent-trial probabilities or sampling uncertainties.

The all-row result is:

$$\langle\Delta\sigma_{\text{res}}\rangle_{\text{all}} = 0.034265 \mu\text{K}, \quad (35)$$

with a full control span

$$\Delta\sigma_{\text{res}}^{\text{span}} = 0.056006 \mu\text{K}. \quad (36)$$

In this grid-weighted descriptive ledger, the Planck baseline is ahead in 99.31% of the rows, but the same audit contains CPTG-ahead rows and a full control span larger than the mean gap. The percentage is not an independent-trial probability. This is why the paper reports a stable near-degeneracy rather than a theory-only residual.

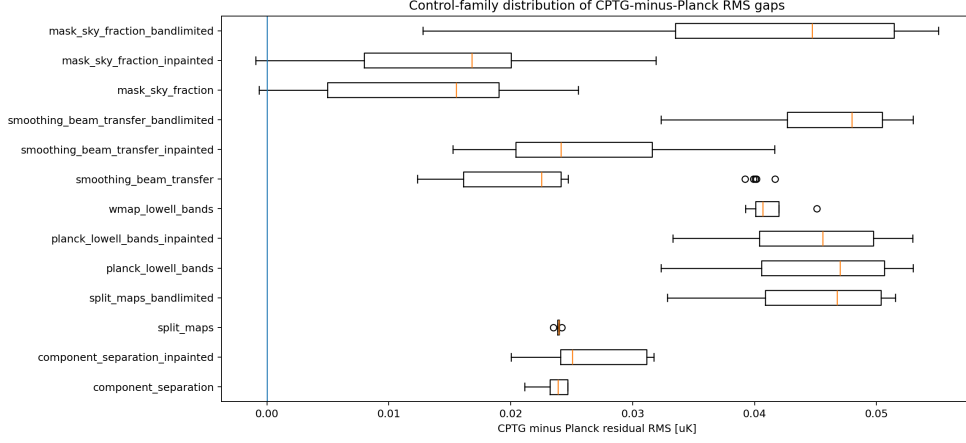


Figure 17: Distribution of CPTG-minus-Planck RMS gaps by control family. The mask and smoothing families have the largest attribution leverage because their scatter is comparable to the mean gap.

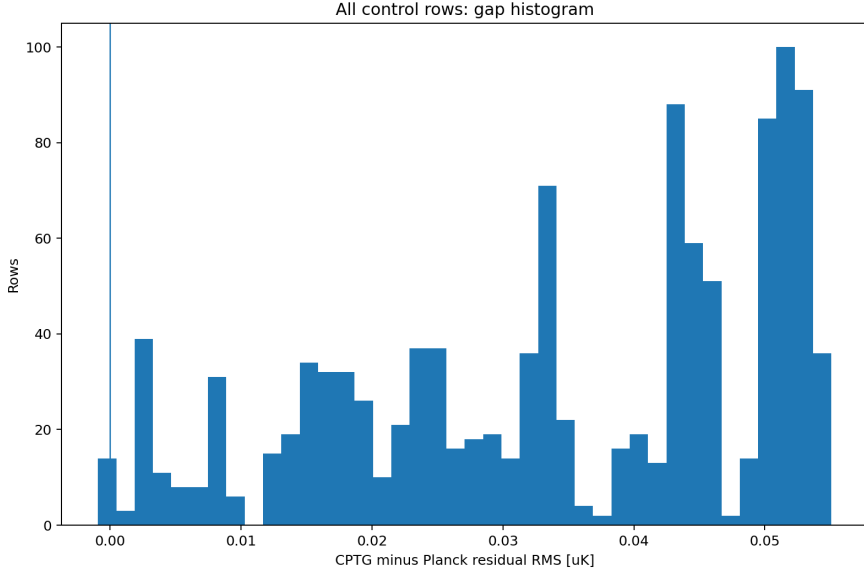


Figure 18: Histogram of all control rows. The distribution is tightly concentrated near zero on an absolute CMB scale, but the small residual crosses zero in high-latitude sky-selection rows.

Control family	Rows	Mean absolute gap [μ K]	Span [μ K]	Ratio	Attribution read
component maps	5	0.023518	0.003541	6.641	descriptive sign-stable; comparison-limited
component maps, inpainted	5	0.026408	0.011678	2.261	descriptive sign-stable; comparison-limited
split maps	5	0.023891	0.000690	34.641	descriptive sign-stable; comparison-limited
split-map bands	20	0.044491	0.018676	2.382	descriptive sign-stable; comparison-limited
Planck low- ℓ	20	0.044683	0.020671	2.162	descriptive sign-stable; comparison-limited
Planck low- ℓ , inpainted	20	0.044103	0.019669	2.242	descriptive sign-stable; comparison-limited
WMAP low- ℓ	4	0.041417	0.005865	7.062	descriptive sign-stable; comparison-limited
smoothing	30	0.023161	0.029317	0.790	descriptive span-comparable; attribution-limited
smoothing, inpainted	30	0.026067	0.026374	0.988	descriptive span-comparable; attribution-limited
smoothing bands	120	0.045343	0.020671	2.194	descriptive sign-stable; comparison-limited
mask / sky fraction	150	0.013305	0.026203	0.508	descriptive sign-changing; attribution-limited
mask / sky fraction, inpainted	150	0.014768	0.032826	0.450	descriptive sign-changing; attribution-limited
mask bands	600	0.042323	0.042286	1.001	descriptive sign-stable; comparison-limited

Table 22: Control-family attribution table. The ratio column is the mean absolute RMS gap divided by the family span. It is a heuristic attribution ratio, not a calibrated probability statistic. Values near or below unity indicate only that the sampled control variation is comparable to the mean gap.

Because the family span is an extreme-range statistic and the control rows are correlated, the mean-gap/span ratio has no direct probability calibration. Its role is descriptive: it identifies families in which the sampled control variation is comparable to the mean differential gap.

The audit therefore strengthens the claim boundary. The comparison-map closure is real in the sense that CPTG stays adjacent to the Planck envelope across many distinct observational control rows. At the same time, the remaining sub- μ K Planck-over-CPTG edge is smaller than the full observational-control span and becomes sign-changing under high-latitude sky cuts. The residual is consequently attribution-limited.

16 Finite null-envelope discrimination control

The null-envelope test addresses the most important possible objection to a phase-locked map-space comparison. Because the observed phase scaffold is shared, one could ask whether phase locking alone would make broad classes of generic angular-power envelopes appear close to the observed map after amplitude-plus-offset fitting. The finite null library tests that possibility by holding the phase scaffold and fitting rule fixed while replacing

the envelope.

The test uses the same observed Planck component maps and the same valid-sky amplitude-plus-offset metric. Only the angular-power envelope is changed. The library includes the locked CPTG envelope, the Planck baseline envelope, the observed self-spectrum control, a smoothed observed-spectrum control, flat C_ℓ and flat D_ℓ nulls, a fitted power-law D_ℓ null, shuffled-spectrum nulls, and small tilt/ripple perturbations around CPTG and Planck. The perturbation rows are diagnostic only; they are not used to retune the locked CPTG result.

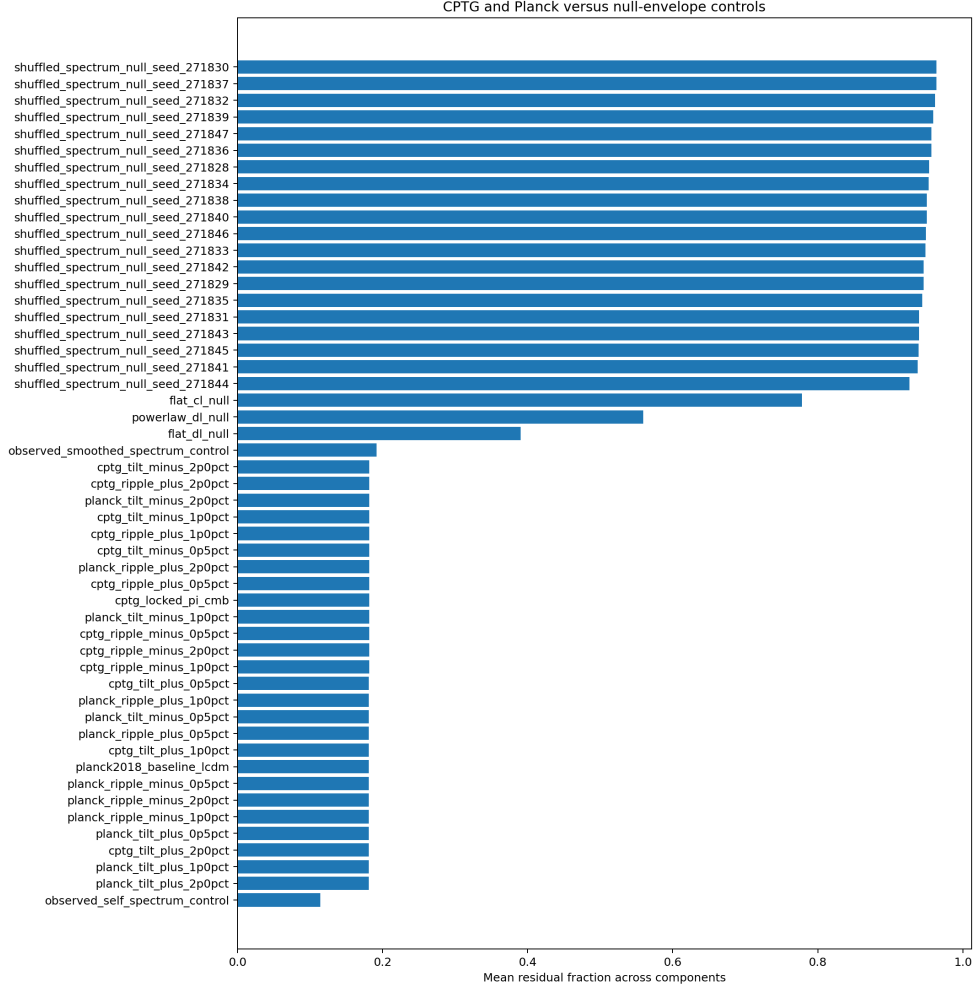


Figure 19: CPTG and Planck compared with null-envelope controls under the same phase-locking and amplitude-plus-offset fitting rule. The tested flat, power-law, and shuffled envelopes fail under the same map-space protocol, showing that shared phases alone are insufficient for these generic null families.

Envelope or family	Mean rank	Mean residual	Mean RMS [μK]	Interpretation
Observed self-spectrum control	1.00	0.114366	11.503	Empirical reconstruction baseline
Planck baseline	9.20	0.181251	18.230	Benchmark envelope
CPTG locked pi-CMB	18.80	0.181485	18.253	Locked CPTG coordinate
Smoothed observed spectrum	28.00	0.191710	19.282	Data-informed smooth control
Flat D_ℓ null	29.00	0.390549	39.281	Generic null fails
Power-law D_ℓ null	30.00	0.559238	56.247	Smooth null fails
Flat C_ℓ null	31.00	0.777863	78.236	Generic null fails
Shuffled-spectrum nulls	41.50	0.949117	95.461	Acoustic structure destroyed

Table 23: Finite null-envelope discrimination summary. Values are means over Planck component maps. The observed self-spectrum is an empirical reconstruction control built directly from the same map-spectrum amplitude; its nonzero residual is treated as a pipeline baseline rather than a strict mathematical lower bound.

The observed self-spectrum control retains a nonzero mean residual RMS of $11.503 \mu\text{K}$ under the finite- ℓ_{max} , masked harmonic reconstruction used here. It is therefore treated as an empirical reconstruction baseline for this pipeline, not as a strict global floor.

The Planck baseline has mean residual RMS $18.230 \mu\text{K}$, while the locked CPTG envelope has mean residual RMS $18.253 \mu\text{K}$. Their mean separation is only

$$18.253220 - 18.229702 = 0.023518 \mu\text{K}. \quad (37)$$

By contrast, the flat D_ℓ , power-law D_ℓ , flat C_ℓ , and shuffled-spectrum nulls are worse by tens of microkelvin. The finite null test therefore shows that the observed phase scaffold alone is insufficient for the tested generic null families; the angular-power envelope remains decisive in the comparison.

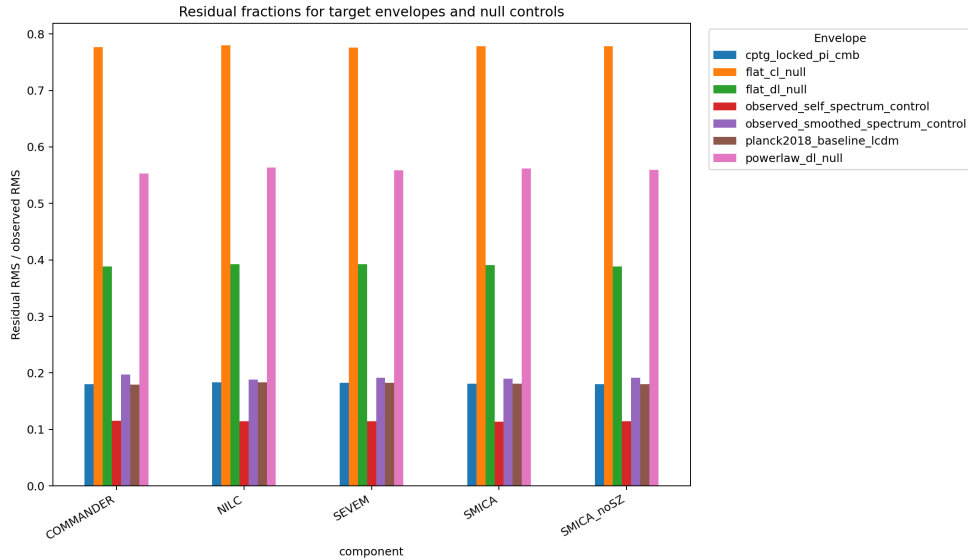


Figure 20: Residual fractions for target envelopes and null controls. CPTG and Planck remain in the high-performing envelope basin, while generic nulls are separated by large residual penalties.

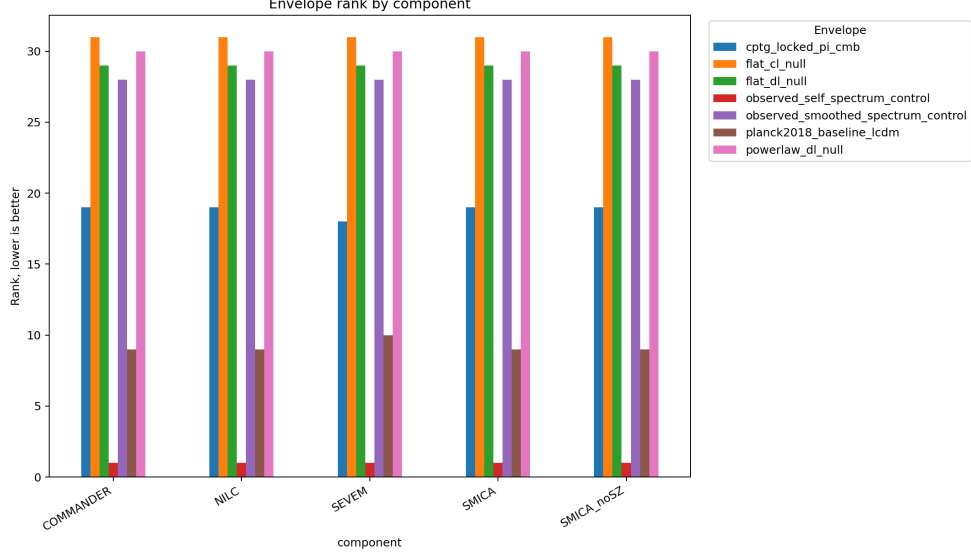


Figure 21: Envelope rank by Planck component map. The data self-spectrum control ranks first in all five retained component rows. Ranks are relative to the finite 51-model library used in this audit and would change if perturbation-library density changed. Planck and CPTG remain near the top of the nontrivial envelopes, while the tested generic null envelopes rank much lower.

Component	Rank _{Planck}	Rank _{CPTG}	RMS _{Planck}	RMS _{CPTG}	$\Delta\sigma_{\text{res}}$	Best null / RMS
Commander	9	19	18.094	18.118	0.023214	flat-dl-null / 39.196
NILC	9	19	18.389	18.414	0.024689	flat-dl-null / 39.346
SEVEM	10	18	18.323	18.345	0.021148	flat-dl-null / 39.447
SMICA	9	19	18.191	18.215	0.023866	flat-dl-null / 39.248
SMICA-noSZ	9	19	18.150	18.174	0.024671	flat-dl-null / 39.165

Table 24: Component-level null-envelope rankings. Ranks are library-relative. The last column lists the best generic null row for that component, demonstrating that the tested generic null families remain far from the CPTG/Planck basin.

The perturbation rows identify a narrow high-performing basin around the Planck and CPTG envelopes. A few small Planck or CPTG tilt perturbations produce marginally smaller residual RMS than the locked rows. This is not a license to retune the CPTG branch. It is a sensitivity diagnostic showing that the map-space metric responds to sub-percent envelope-shape perturbations around the high-performing basin.

Model	Family	Mean rank	Mean RMS [μK]	Mean correlation
planck-tilt-plus-2p0%	planck-perturbation	2.00	18.189	0.983511
planck-tilt-plus-1p0%	planck-perturbation	3.00	18.208	0.983475
cptg-tilt-plus-2p0%	cptg-perturbation	4.20	18.215	0.983462
planck-tilt-plus-0p5%	planck-perturbation	4.80	18.219	0.983456
planck-ripple-minus-1p0%	planck-perturbation	6.40	18.224	0.983447
planck-ripple-minus-2p0%	planck-perturbation	6.60	18.224	0.983446
planck-ripple-minus-0p5%	planck-perturbation	8.00	18.226	0.983443
planck2018-baseline-lcdm	benchmark	9.20	18.230	0.983436
cptg-tilt-plus-1p0%	cptg-perturbation	10.00	18.233	0.983429
planck-ripple-plus-0p5%	planck-perturbation	10.80	18.235	0.983426
planck-tilt-minus-0p5%	planck-perturbation	12.60	18.241	0.983415
planck-ripple-plus-1p0%	planck-perturbation	12.80	18.242	0.983414

Table 25: Top non-self-spectrum envelope rows in the null-envelope audit. Perturbation rows are diagnostic only. Mean ranks are relative to the finite 51-model library used here. The rows show the local response of the high-performing envelope basin and are not used to redefine the locked CPTG row.

The null-envelope control is therefore the finite discrimination layer of the paper. Sharing the observed phase scaffold is not enough: the angular-power envelope still has to carry the required acoustic and large-angle structure. The locked CPTG envelope remains adjacent to the Planck benchmark while the tested generic null families fail, without implying uniqueness over all possible envelopes.

17 Evidence synthesis

The map-space evidence chain can now be read as a sequence of increasingly restrictive checks. The visual SMICA comparison demonstrates that the locked CPTG envelope projects into the same large-scale temperature morphology as the observed map when the phase scaffold is shared. The component-map and split-map tests then show that this closure is not restricted to one Planck component product or one internal map split. The smoothing and sky-fraction ladders identify where the remaining small residual is observationally sensitive. The control-scatter audit then compares the residual gap to the descriptive variation across those controls. Finally, the null-envelope audit asks whether phase locking alone is enough to make the tested generic null envelopes look successful. It is not.

The evidence structure is summarized in Table 26. The table is intentionally written as a claim ledger rather than as a single combined score. A comparison-map closure result is strongest when each evidence layer is labeled by its data object, metric, and claim boundary.

Evidence layer	Data or control object	Main numerical result	Interpretation
Visual SMICA map closure	Planck SMICA I_STOKES with TMASK at $N_{\text{side}} = 256$	fitted residual RMS $18.218928 \mu\text{K}$, correlation 0.983434	CPTG and observed map share the same phase scaffold; residual map is the meaningful diagnostic.
Planck component maps	SMICA, SMICA-noSZ, NILC, SEVEM, Commander	mean full-band RMS gap $0.023518 \mu\text{K}$	CPTG remains adjacent to the Planck benchmark across distinct component-separation products.
Planck split maps	SMICA split products	mean RMS gap $0.023891 \mu\text{K}$, split span $0.000690 \mu\text{K}$	Closure is very stable under internal Planck split-map replacement.
Smoothing ladder	native through 120 arcmin Gaussian smoothing	minimum gap near 20–40 arcmin; growth at 120 arcmin	The gap is smoothing- and scale-sensitive; beam/transfer and envelope-shape attribution remains unresolved.
Mask and sky-fraction ladder	TMASK thresholds and Galactic latitude cuts	no-cut gap $0.02379 \mu\text{K}$, $ b > 40^\circ$ gap $0.00146 \mu\text{K}$	The small gap shrinks strongly at high Galactic latitude and can become sign-changing in individual rows.
Control-scatter audit	1159 combined control rows	mean gap $0.034265 \mu\text{K}$, span $0.056006 \mu\text{K}$	Descriptive control-grid mean; the full span exceeds the mean gap.
Finite null-envelope discrimination	flat, power-law, smoothed, shuffled, and perturbed envelopes	Planck $18.230 \mu\text{K}$, CPTG $18.253 \mu\text{K}$, shuffled nulls $95.461 \mu\text{K}$	Shared phases alone are insufficient for the tested generic null families.

Table 26: Final evidence chain for the CMB comparison-map closure paper. The table separates primary map-space closure rows from attribution and discrimination controls.

The central result is therefore not that CPTG exactly equals the Planck baseline in every possible statistic. The result is that one locked CPTG geometric-pi construction can be translated into the measured CMB harmonic field coordinate, remain in the same high-performing reconstructed-field basin as the Planck baseline across a large number of observational controls, and reject broad generic envelopes under the same translation.

18 Claim-boundary and publication-scope discipline

The result requires careful language. The comparison-map protocol is powerful precisely because the geometric row is locked, the observational phase coordinate is declared, and the same residual metric is applied to CPTG and Planck. It is not a full Planck likelihood analysis, and it is not a license to tune the CPTG row after seeing the residuals.

Table 27 states the final claim boundary used by this paper.

Layer	Status in this paper	Proper claim	Not claimed
Observed field-coordinate comparison	Primary map-space evidence	The locked CPTG geometry translates into the SMICA harmonic field coordinate and reconstructs map space with a small residual field and high correlation.	Independent spectrum-space or likelihood-grade closure.
Planck component maps	Primary robustness evidence	Closure is stable across major Planck temperature component products.	Proof that all Planck likelihood nuisance structure has been reproduced.
Planck split maps and WMAP low- ℓ bands	Primary robustness evidence	Closure survives internal split-map and cross-mission low-multipole checks.	Full cross-mission likelihood validation.
Smoothing and mask ladders	Attribution controls	The remaining gap is smoothing-, mask-, and sky-selection-sensitive; attribution among transfer and envelope-shape effects remains limited.	Independent new CPTG validation layer.
Control-scatter audit	Attribution control	The descriptive grid mean is smaller than the full control span and is therefore attribution-limited.	A probability statement from independent trials or a proof that the Planck-over-CPTG edge is physically meaningless.
Null-envelope audit	Finite discrimination control	Phase locking alone does not make the tested generic null envelopes successful; CPTG remains adjacent to Planck while those controls fail.	An exhaustive theorem over all possible envelopes or a full likelihood/posterior replacement for Planck.
TT/TE/EE spectrum-space closure	Future separate paper	Disclosed as continuing work outside this comparison-map paper.	Primary evidence for this paper.
Calibration-stress or response rows	Future diagnostic protocol	Useful only when reported separately and applied symmetrically under a declared protocol.	A way to rescue or redefine the primary result after the fact.

Table 27: Final claim-boundary table. Rows marked primary are used directly in the paper’s main claim. Rows marked attribution or discrimination support the interpretation but do not redefine the primary result.

The paper therefore makes a narrow but strong statement:

$$\begin{aligned} &\text{CPTG achieves real CMB comparison-map closure} \\ &\text{in a locked pi-branch coordinate.} \end{aligned} \tag{38}$$

It does not make the stronger statement:

$$\text{CPTG has completed all Planck likelihood and polarization tests.} \tag{39}$$

19 Implications

The first implication is methodological. A pi-based comparison is not the same object as a Planck-style optimized likelihood. Planck begins from a likelihood architecture in which cosmological parameters, foreground terms, calibration-like factors, beam corrections, and other nuisance quantities are profiled or marginalized inside a mature observational pipeline. The CPTG comparison-map test instead begins with a locked geometric row and

asks whether that row can survive translation into a declared observational coordinate without theory-side retuning.

The second implication is representational. The result is stronger than a resemblance between two temperature images. A CMB map is the real-space representation of angular structure distributed across the spherical-harmonic field. In this test the fixed CPTG envelope is carried through that field coordinate before the inverse transform is performed. Closure therefore means that the geometric construction survives translation across the retained angular organization of the observed CMB, not merely that it produces a visually similar heat map.

The third implication is physical but still bounded. The locked CPTG CMB row is not an arbitrary smooth spectrum. If it were, the null-envelope controls would have produced similar map-space performance for flat, power-law, or shuffled spectra. They did not. The CPTG envelope carries enough acoustic and large-angle structure to remain essentially adjacent to the Planck benchmark in the phase-locked field-coordinate basin.

The fourth implication concerns residual attribution. The Planck baseline remains slightly ahead in most RMS comparisons, but the difference is at the level of hundredths of a microkelvin in the map-space RMS gap. The smoothing and mask ladders show that this edge changes with beam-scale treatment and sky selection. The high-latitude rows are especially important because the gap shrinks toward zero and sometimes changes sign. Therefore the residual should not be assigned uniquely to CPTG. It is sensitive to map construction, transfer response, sky fraction, foreground cleaning, comparison-coordinate projection, and the fact that the Planck baseline is optimized in its native framework; contributions from those layers cannot be uniquely separated by the present map-space controls.

The fifth implication is that the comparison-map closure result is useful even before a full likelihood-grade CMB program is finished. It gives CPTG a reproducible field-facing audit layer: real maps, declared files, declared masks, declared scripts, declared output tables, and a clear primary-versus-diagnostic claim structure.

20 Future work outside this paper

The immediate future work is a separate TT/TE/EE spectrum-space paper. That paper should not be merged into the present comparison-map closure paper because it answers a different question. The map paper asks whether the locked CPTG envelope projects into real CMB map coordinates and remains adjacent to the Planck envelope under real-map controls. The spectrum paper asks whether the same locked row closes directly against binned angular spectra without using the observed phase scaffold.

A second future work package is likelihood-grade Planck validation. That requires a declared high- ℓ TTTEEE likelihood protocol, official or pipeline-quality nuisance treatment, symmetric Planck-control and CPTG rows, and calibration-stress diagnostics kept separate from the primary likelihood rows unless those stress constraints are included from the

start. The required workflow is:

$$\begin{aligned}
&\text{declare likelihood protocol} \longrightarrow \text{lock CPTG and control rows} \\
&\longrightarrow \text{profile nuisance block symmetrically} \\
&\longrightarrow \text{report primary and diagnostic tables separately.}
\end{aligned} \tag{40}$$

A third future work package is independent reproduction. The files and commands in the appendix are included so that the comparison-map result can be rerun outside this conversation. Independent reproduction should begin from the same map products and should preserve the distinction among primary map-space evidence, attribution controls, null-envelope controls, and future spectrum-space tests.

21 Conclusion

This paper constructs a CMB field-coordinate closure test for the locked CPTG geometric-pi CMB comparison coordinate using real Planck and WMAP temperature products. The observed phase scaffold defines the measured harmonic coordinate; the locked CPTG envelope and Planck benchmark are translated through the same phase-locked rule, reconstructed into real map space, and compared with the same amplitude-plus-offset residual metric.

The result is nontrivial. In the primary SMICA visual comparison, the fitted CPTG map reaches a residual RMS of $18.218928 \mu\text{K}$ with observed-versus-fitted correlation 0.983434 . Across Planck component maps, the mean full-band CPTG-minus-Planck RMS gap is $0.023518 \mu\text{K}$. Across the Planck SMICA full and split products, the mean full-band gap is $0.023891 \mu\text{K}$ with a full-and-split span of only $0.000690 \mu\text{K}$. Across the 1159 nested and correlated control rows in the control-scatter audit, the descriptive grid mean is $0.034265 \mu\text{K}$, while the full control span is $0.056006 \mu\text{K}$.

The null-envelope audit shows why this matters. The same observational field coordinate does not rescue the tested generic null envelopes. Flat, power-law, and shuffled spectra perform substantially worse, while the locked CPTG row remains adjacent to the Planck benchmark. The closure is therefore not a heat-map resemblance or an automatic consequence of sharing observational phases; it depends on the angular-power structure carried by the locked geometric-pi translation.

The disciplined conclusion is that CPTG achieves a strong CMB field-coordinate closure result, demonstrated through real comparison maps. The remaining Planck-over-CPTG edge is small, measurable, and attribution-limited. It should be reported honestly, not hidden; but it should also not be overinterpreted as a unique CPTG physical residual without full likelihood-grade nuisance, calibration, beam, foreground, and spectrum-space treatment. TT/TE/EE spectrum closure and Planck likelihood-grade validation are continuing work for separate papers.

22 Data and code availability boundary

The manuscript package contains the TeX source, compiled preview PDF, figures, CSV result tables, command ledger, analysis scripts, package hashes, and reproducibility notes used to preserve the reported comparison-map rows. The large external CMB survey products are not redistributed inside the manuscript package. Reproduction therefore requires downloading the Planck and WMAP files listed in Tables 3 and 4, placing them in the local working directory, and running the commands in Appendix A. Exact SHA-256 hashes of those external FITS inputs were not retained in the original run record and are not fabricated in this release. This boundary keeps the paper package lightweight while preserving the retained analysis recipe and its documented limitations.

The analysis scripts are not additional theory assumptions. They implement the declared comparison-coordinate protocol: read a real CMB map, build the observed phase scaffold, apply the locked CPTG and Planck envelopes through the same transport rule, fit amplitude and offset in the same way, and report the residual metrics. Any independent reproduction should preserve this order of operations and should keep primary map-space rows separate from attribution and diagnostic rows.

A Reproducibility command ledger

The following commands record the working command sequence used for the map-space closure paper. They assume a local working directory such as `D`:

Downloads, with the Planck and WMAP map files listed in Tables 3 and 4 already present. The commands are written with bare script names, matching the original working directory; when running directly from the packaged release root, prefix each script name with `scripts`

. The commands are not part of the statistical claim by themselves; they are included so the analysis path can be audited.

A.1 Environment

The working environment used Python with `healpy`, `numpy`, `scipy`, `astropy`, `matplotlib`, `pandas`, and `camb`. The tested local environment was named `cptg-cmb`. Exact package-version pins were not retained in the original evidence record and are therefore not reconstructed or invented here; the release package records that limitation explicitly. The required analysis resolution in all map-space rows was $N_{\text{side}} = 256$.

A.2 Primary visual comparison

```
python cptg_cmb_visual_compare_v2.py ^
--cmb-map COM_CMB_IQU-smica_2048_R3.00_full.fits ^
--nside 256 ^
```

```
--out cptg_cmb_visual_compare_v2
```

A.3 Planck component and inpainted controls

```
python cptg_planck_component_robustness_v4.py ^  
--nside 256 ^  
--out cptg_planck_component_robustness_v4  
python cptg_planck_component_robustness_v4.py ^  
--nside 256 ^  
--use-inpainted ^  
--out cptg_planck_component_robustness_v4_inpainted
```

A.4 Planck split-map stability

```
python cptg_planck_splitmap_stability.py ^  
--nside 256 ^  
--out cptg_planck_splitmap_stability  
python cptg_planck_splitmap_stability.py ^  
--nside 256 ^  
--bands 2-30,2-40,2-64,2-95 ^  
--out cptg_planck_splitmap_stability_bands
```

A.5 Smoothing and mask ladders

```
python cptg_cmb_smoothing_ladder_v2.py ^  
--nside 256 ^  
--out cptg_cmb_smoothing_ladder_v2  
python cptg_cmb_smoothing_ladder_v2.py ^  
--nside 256 ^  
--use-inpainted ^  
--out cptg_cmb_smoothing_ladder_v2_inpainted  
python cptg_cmb_smoothing_ladder_v2.py ^  
--nside 256 ^  
--bands 2-30,2-40,2-64,2-95 ^  
--out cptg_cmb_smoothing_ladder_v2_bands  
python cptg_cmb_mask_ladder.py ^  
--nside 256 ^  
--out cptg_cmb_mask_ladder  
python cptg_cmb_mask_ladder.py ^  
--nside 256 ^  
--use-inpainted ^
```

```
--out cptg_cmb_mask_ladder_inpainted
python cptg_cmb_mask_ladder.py ^
--nside 256 ^
--bands 2-30,2-40,2-64,2-95 ^
--out cptg_cmb_mask_ladder_bands
```

A.6 WMAP low-multipole comparison

```
python cptg_wmap_lowell_bandtest.py ^
--wmap wmap_ilc_9yr_v5.fits ^
--bands 2-30,2-40,2-64,2-95 ^
--out cptg_wmap_lowell_bandtest
```

A.7 Control-scatter and null-envelope audits

```
python cptg_cmb_control_scatter_audit.py ^
--out cptg_cmb_control_scatter_audit
python cptg_cmb_null_envelope_controls.py ^
--nside 256 ^
--out cptg_cmb_null_envelope_controls
```

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